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Automation of the characterization of superconducting transmon qubits

Semester Thesis

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May 25, 2023

Abstract

In order to use superconducting circuits for quantum experiments they need to be of high quality, which can be assessed by characterizing the circuits main physical properties. We successfully developed a software to automate this characterization of superconducting transmon qubits that are coupled dispersively to a microwave cavity. We designed a general user interface that allows for complete and easy control over the parameters used during the characterization procedure. For the analysis of the measurement data generated during the procedure we developed tools to fit the data, identify features in it and make decisions based on the analysis' results using a peak detection algorithm, the coefficient of determination and a Savitzky-Golay filter. For the presentation of the software's output we developed a log file that documents the measurements, analysis results and decisions that were made during the characterization procedure and a PDF report file summarizing the cavity's and qubit's properties that were determined in the characterization. The software is compatible with Python 2.7 up to current versions. It was tested successfully on five qubits.

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Introduction

For more than five decades the number of transistors on a computer chip has doubled roughly every two years which has lead to an exponential increase of computational power. With transistor size reaching the scale of a few atoms this rule of thumb known as Moore's law is bound to come to an end in the near future. Therefore the search for new methods of computing attracts more and more interest [1]. Quantum computers are one emerging technology that promises the next, great leap in computational power. Among the many different platforms that are explored as potential architectures to realize quantum computers superconducting circuits are one of the most promising candidates. They benefit greatly from the continuous improvements in the technologies that enabled the production of ever smaller silicon transistors and that are also used in the fabrication of superconducting circuits. Recent advancements have demonstrated superconducting qubits that display lifetimes of the order of hundreds of microseconds up to milliseconds and gates with fidelities larger than 99 % operating on time scales of the order of hundreds of nanoseconds [2].

Superconducting (SC) circuits are, however, at their current state not suited to store quantum information over a time period on the order of seconds. One way of overcoming this issue would be to combine superconducting circuits with other platforms such as mechanical resonators into hybrid quantum systems. These offer the possibility to utilize their component's individual strengths such as the long lived nature of mechanical resonator states and short gate times of superconducting circuits in the development of novel devices like a quantum memory. Furthermore, interfacing different physical platforms can give rise to new, interesting, physical phenomena that allow us to study fundamental physics. By combining a transmon, which is a specific type of superconducting qubit, with a high-overtone bulk acoustic-wave resonator (HBAR) into what is called a $\hbar$ BAR, researchers were able to prepare and control superposition states of a 16 μg resonator [3]. This is so far the most massive object that has been observed to be in a superposition state, shedding new light onto the boundary between the quantum and classical world. A picture of a similar device can be found in figure 1.1.

In the displayed $\hbar$ BAR an electric field will build up between the two large pads of the transmon qubit. By bonding the transmon qubit to an HBAR chip this electric field is disturbed which drastically reduces the qubit's lifetime. For this reason it is important to choose qubits with high lifetimes during fabrication of the hybrid system in order to be able to operate the final device. This can only be ensured by testing each individual qubit and measuring its lifetime, which can be a time consuming process.

The process of characterizing the qubit's lifetime together with other important pa-

1. Introduction

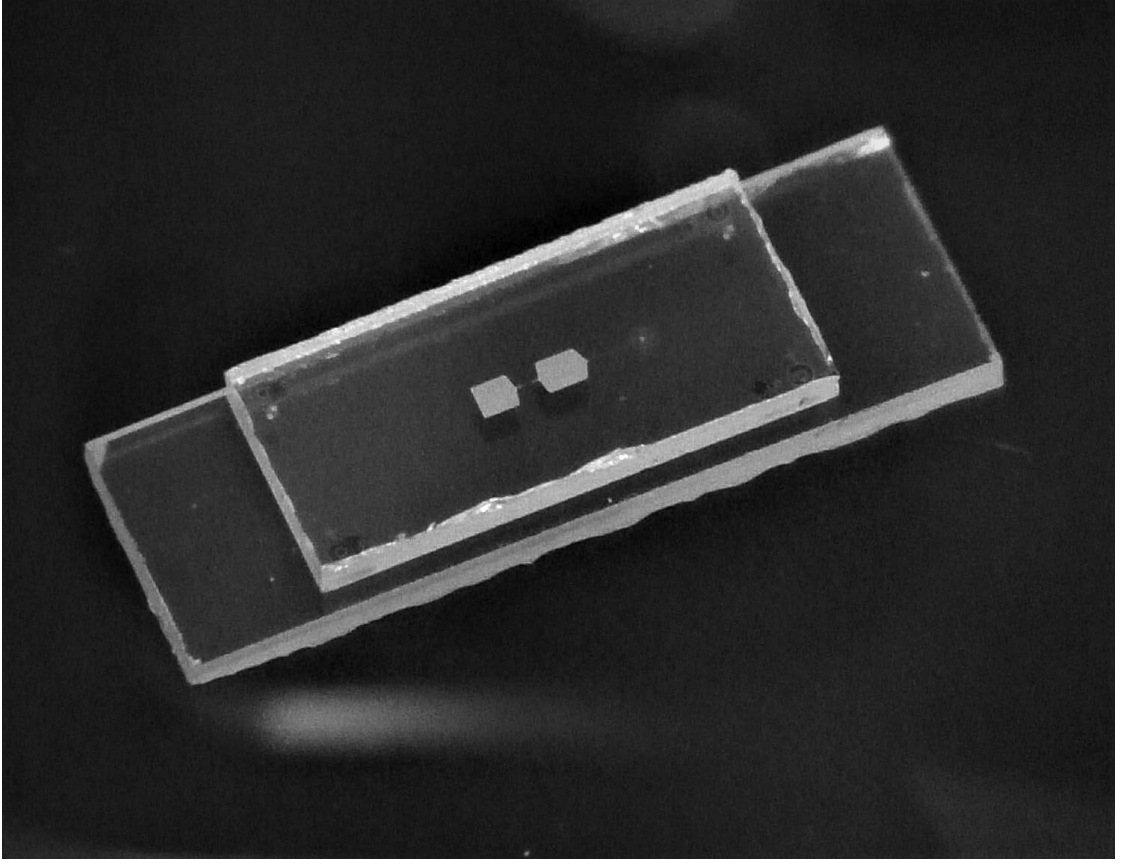


Figure 1.1.: Image of a transmon chip bonded to a HBAR chip.

rameters involves standardized measurements that follow a specific order. Therefore this task seems perfect for automation, which would free up a lot of time that could instead be used for designing and fabricating new, improved qubits. Furthermore automation would allow for overnight testing which would maximise the time usage of the measurement infrastructure. In the end this could lead to shorter qubit design, fabrication and testing cycles and therefore result in faster improvements of qubit quality.

In this report we first give an introduction into the physics behind transmon qubits and how they can be controlled and measured using microwave resonators and pulses in section 2. In section 3 we discuss the experimental realization of these measurements. The specific order of measurements that the characterization has to follow together with the necessary analysis steps during the procedure are discussed in detail in section 4. We present the results of automating the characterization procedure in section 5, followed by a discussion of the limitations that our approach is subject to and potential ways of overcoming these in the future in section 6. We conclude the report with a short summary in section 7.

Circuit quantum electrodynamics: measurements of superconducting qubits

2.1. Superconducting Transmon Qubit

A superconducting transmon qubit consists of two superconducting islands that are shunted capacitively and that are connected by a tunneling junction, called Josephson junction. A schematic drawing of this circuit can be seen in figure 2.1a. This circuit is similar to that of a LC resonant circuit, where the Josephson junction would be replaced by a linear inductor. The Hamiltonian of the LC resonant circuit is equivalent to that of a quantum harmonic oscillator. In the case of the transmon circuit the Josephson junction acts as a non-linear inductor. This non-linearity of the Josephson junction is crucial as it will introduce anharmonicity into the transmon's energy spectrum. Whereas the harmonic oscillator energy levels are equidistantly spaced, the transmon's energy level separation decreases with increasing energy, as shown in figure 2.1b. The anharmonicity can be quantified as $\alpha = \omega_{ge} - \omega_{ef}$ where ω_{ge} denotes the frequency difference between the transmon's ground and first excited state and ω_{ef} denotes the frequency difference between the transmon's first and second excited state. The difference in the transition energies between neighbouring energy levels makes it possible to address individual transitions. This can be achieved by tuning the frequency of the microwave pulses used to drive and control the qubit to match the energy of the specific transition of interest. For the moment, we can therefore focus on the ground and first excited state of the transmon, which we denote by $|g\rangle$ and $|e\rangle$, respectively. This reduces the transmon's energy spectrum to a two-level-system (TLS), as can be seen in figure 2.1. In the following we will usually associate the energy difference between the ground and first excited state with the frequency $\omega_a = \omega_{ge}$, which we will call the qubit (resonance) frequency. Treating the transmon qubit as a two-level-system, allows us to write down its Hamiltonian as

$$\hat{H}_{TLS} = \frac{1}{2} \hbar \omega_a \hat{\sigma}_z \quad (2.1)$$

where $\hat{\sigma}_z = |e\rangle\langle e| - |g\rangle\langle g|$ is the Pauli z-matrix [4], [5].

The qubits that are used in our lab are made by depositing aluminium on a sapphire substrate. The Josephson junction is formed by an aluminium-oxide layer between two

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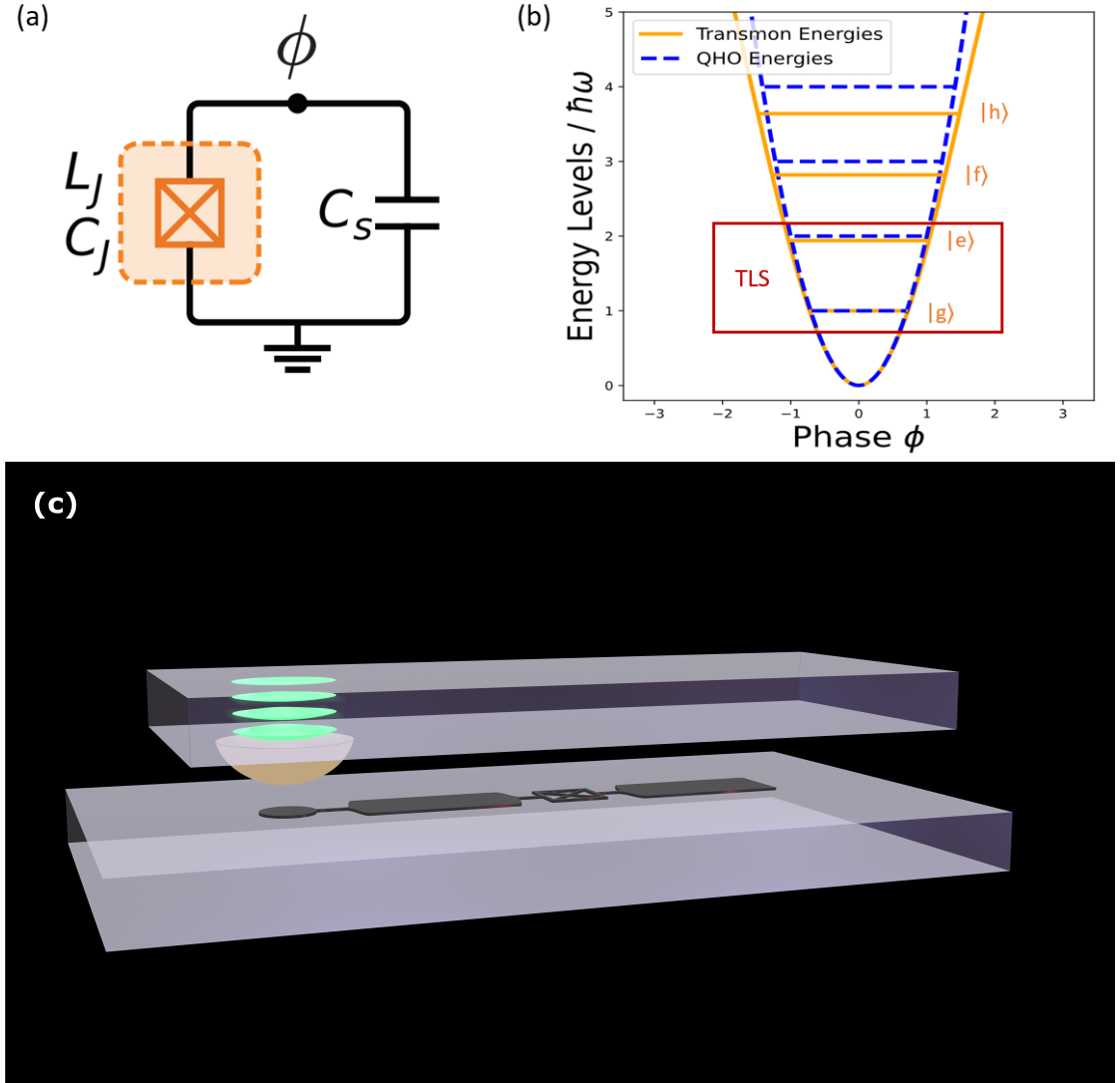


Figure 2.1.: (a) Circuit of a superconducting transmon qubit. The figure was taken from [4]. (b) Energy diagram of transmon qubit in orange and of a quantum harmonic oscillator in blue. The two level system that is used as the qubit is circled in red. The figure was taken from [6] and adapted. (c) Schematic drawing of a transmon qubit bonded to a HBAR. The qubit chip can be seen on the bottom. The transmon consists of two large pads that are responsible for the capacitive coupling. They are separated by a Josephson junction. The coupling to the HBAR chip on top is achieved via the antenna. A picture of the fabricated device can be found in figure 1.1.

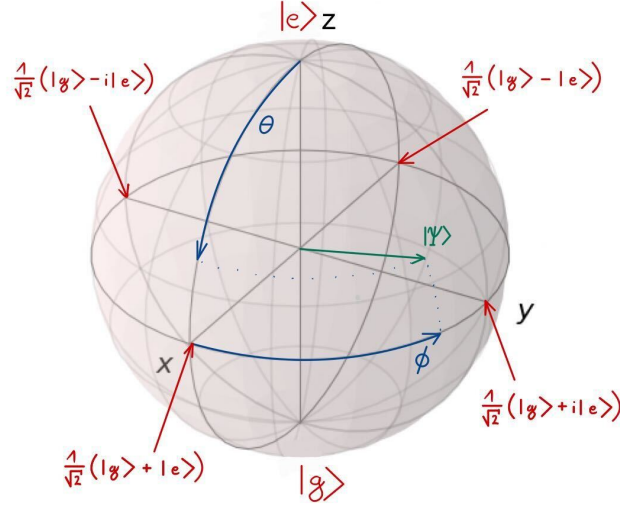


Figure 2.2.: Bloch sphere representation of the qubit state.

layers of aluminium. A schematic drawing of the layout of the transmon is shown in figure 2.1c. Since transmon qubits rely on the Josephson tunneling junction, they need to be operated below the critical temperature of aluminium, which lies at roughly 1.2 K [7].

2.1.1. Bloch sphere

A pure state of any quantum TLS can always be written as $|\psi\rangle = a|g\rangle + b|e\rangle$ where the complex coefficients a, b fulfill the normalization condition $|a|^2 + |b|^2 = 1$. Since an overall phase of the state is not physically observable we can ignore it and only consider the relative phase between the two coefficients. Therefore we can assume a to always be real which enables us to write the qubit state as

$$|\psi\rangle = \sin\frac{\theta}{2}|g\rangle + e^{i\phi}\cos\frac{\theta}{2}|e\rangle \quad (2.2)$$

where $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq 2\pi$. Interpreting θ as the polar and ϕ as the azimuthal angle the state can be visualized as a vector of unit length, the Bloch vector, on the unit sphere, Bloch sphere. The north pole of this sphere represents the excited state, whereas the south pole represents the ground state. All other points on the sphere are superpositions of the qubit's ground and excited state. A picture of the Bloch sphere and its most important points can be seen in figure 2.2. We will use this visualization of the qubit state later when talking about specific qubit operations. First, let us talk about how to control a SC qubit in general.

2.2. Jaynes-Cummings Hamiltonian

The qubit is controlled and read out by coupling it to a microwave cavity. The isolated cavity without a qubit inside can be described by the Hamiltonian of a harmonic oscillator

$$\hat{H}_r = \hbar\omega_r \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad (2.3)$$

where $\hat{a}$ and $\hat{a}^\dagger$ are the annihilation and creation operator, respectively. Here we assumed that there is only one relevant cavity mode of frequency ω_r . We will refer to this frequency as the cavity (resonance) frequency.

By applying the rotating wave approximation the coupled qubit-cavity-system can be described by the Jaynes-Cummings-Hamiltonian

$$\hat{H} = \hbar\omega_r \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) + \frac{1}{2} \hbar\omega_a \hat{\sigma}_z + \hbar g \left(\hat{\sigma}_+ \hat{a} + \hat{\sigma}_- \hat{a}^\dagger \right) \quad (2.4)$$

where g is the coupling strength and $\hat{\sigma}_+ = |e\rangle\langle g|$ and $\hat{\sigma}_- = |g\rangle\langle e|$ denote the raising and lowering operator of the TLS. The first two terms of equation 2.4 represent the Hamiltonians of the isolated cavity and qubit as can be seen from equations 2.1 and 2.3. The third term describes the interaction between both systems, which can be interpreted as the exchange of excitations between the two systems [4], [8].

2.2.1. Dispersive limit and qubit readout

The detuning between qubit and cavity is defined as the difference between the qubit and cavity resonance $\Delta = \omega_a - \omega_r$. Our devices are typically operated in a regime where Δ is negative.

Furthermore, operating in the dispersive regime where $|\Delta| \gg g$ allows for quantum non-demolition (QND) measurements of the qubit. By performing a unitary transformation on the Jaynes-Cummings-Hamiltonian given in equation 2.4 and expanding the result up to second order in g , we obtain the dispersive Hamiltonian

$$\hat{H}_{dis} = \hbar \left(\omega_r + \frac{g^2}{\Delta} \hat{\sigma}_z \right) \hat{a}^\dagger \hat{a} + \frac{\hbar}{2} \left(\omega_a + \frac{g^2}{\Delta} \right) \hat{\sigma}_z. \quad (2.5)$$

It is obvious that the Fock states $|n, g\rangle = |n\rangle \otimes |g\rangle$ and $|n, e\rangle = |n\rangle \otimes |e\rangle$ are eigenstates of this Hamiltonian, where $|n\rangle$ is an eigenstate of the harmonic oscillator (equation 2.3) corresponding to n photons being inside the cavity. Therefore we can interpret the result of the interaction between qubit and cavity as a shift of the qubit resonance frequency $\omega_a \rightarrow \omega_a + \frac{g^2}{\Delta}$ and of the cavity resonance frequency $\omega_r \rightarrow \omega_r \pm \frac{g^2}{\Delta}$, which depends now on whether the qubit is in its excited or ground state [8]. This shift of the cavity frequency is called dispersive shift $\chi = \frac{g^2}{\Delta}$ and can be used to measure the qubit's state as will be explained later in this chapter. The effect of the dispersive shift on the transmission spectrum is visualized in figure 2.3: Shown in blue is the transmission spectrum of a

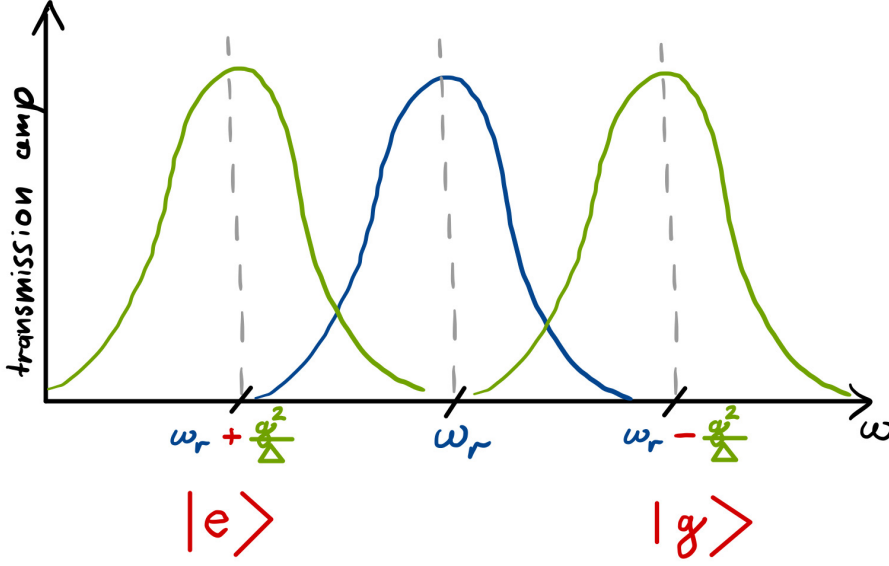


Figure 2.3.: Transmission spectrum of the bare cavity around its resonance frequency ω_r in blue. Transmission spectrum of the cavity interacting with the qubit including the dispersive shift of the cavity's resonance frequency towards (away) from the qubit resonance frequency for the qubit being in the excited (ground) state in green.

bare cavity, which displays a Lorentzian lineshape with central frequency ω_r . Coupling of the cavity to the continuum results in a finite lifetime of the photons inside the cavity and thereby in a finite linewidth of the transmission spectrum. In green one can see that the transmission spectrum is shifted by $\pm\chi$ in frequency depending on the state of the qubit. Since Δ can be assumed to be negative, the transmission spectrum is shifted towards higher (lower) frequencies for the qubit being in its ground (excited) state. This can be seen as a push (pull) on the cavity resonance frequency away from (towards) the qubit's resonance frequency.

This dispersive shift can be used to determine the qubit's state. A schematic representation of this process can be found in figure 2.4. By first sending a control pulse into the cavity the qubit is manipulated and prepared in a certain state. More details on this will be explained in chapter 2.2.2. Afterwards, a readout pulse will be sent through the cavity. The spectrum of this square pulse will be sharply peaked around a central frequency such that the pulse can be treated as a continuous wave at that frequency. The frequency is tuned to match the cavity's resonance frequency for when the qubit would be in its ground state. This makes it possible to distinguish between the two qubit states by looking at the transmitted signal's amplitude. If the qubit is in the ground state, the readout pulse will be in resonance with the cavity and therefore the transmitted signal will be high in amplitude. If on the other hand the qubit is in its excited state, the cavity's resonance frequency will be shifted towards lower frequencies. Therefore the readout pulse and the cavity will be out of resonance and the transmitted signal will

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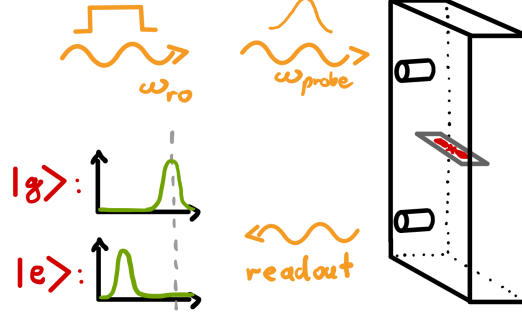


Figure 2.4.: Schematic representation of a dispersive QND measurement.

be low in amplitude. Instead of measuring the amplitude of the transmission spectrum one could also measure the phase of the transmission through the cavity or the phase or amplitude of the reflected signal [8].

The measurement described above will collapse the qubit's state into either the ground or the excited state and is therefore a projective measurement. This projection onto the z-axis becomes more apparent when visualizing the qubit state on the Bloch sphere. Since the qubit state is not destroyed by the act of measuring and remains in the ground or excited state afterwards, such a measurement is called quantum non-demolition measurement.

2.2.2. Coherent Control of qubits in cQED

As discussed in the previous section we can apply microwave pulses around the bare cavity's resonance frequency to perform QND readout of the qubit state. In the dispersive limit these pulses will be far detuned from the qubit resonance frequency. In the following we will examine what happens if we apply microwave pulses around the qubit resonance frequency to our system.

The drive of frequency ω_{dr} on the system can be modelled as

$$\hat{H}_{dr}(t) = \hbar\epsilon(t) \left(\hat{a}^\dagger e^{-i\omega_{dr}t} + \hat{a} e^{+i\omega_{dr}t} \right) \quad (2.6)$$

where $\epsilon(t)$ is proportional to the amplitude of the microwave drive. Adding this term to the Jaynes-Cummings-Hamiltonian given in equation 2.4 and performing a unitary transformation results in

$$\hat{H}_{drive} = \frac{\hbar}{2} \left[\omega_a + 2\frac{g^2}{\Delta} \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) - \omega_{dr} \right] \hat{\sigma}_z + \hbar \frac{g\epsilon(t)}{\Delta} \hat{\sigma}_x + \hbar(\omega_r - \omega_{dr}) \hat{a}^\dagger \hat{a} + \hbar\epsilon(t) \left(\hat{a}^\dagger + \hat{a} \right). \quad (2.7)$$

We will ignore the last two terms of this equation as they only act on the cavity. By choosing $\omega_{dr} = \omega_a + 2\frac{g^2}{\Delta} \left(n + \frac{1}{2} \right)$ the first term will vanish and only the term containing

$\hat{\sigma}_x$ will be left. A microwave pulse at this frequency will therefore generate rotations of the qubit state around the x-axis on the Bloch-sphere. The form of the drive given in equation 2.6 implicitly assumes $\epsilon(t)$ to be real. If we instead choose $\epsilon(t)$ to be purely imaginary and repeat the analysis described above we would obtain a Hamiltonian that only contains a $\hat{\sigma}_y$ -term. Adding a phase of $\frac{\pi}{2}$ to the microwave drive pulse will therefore change the axis of rotation, around which the qubit state is rotated, from the x-axis to the y-axis. Allowing for an arbitrary phase of $\epsilon(t)$ the axis of rotation can be rotated freely in the x-y-plane of the Bloch sphere. Thus, controlling the phase together with the shape and amplitude of the drive pulse allows us to reach any point on the Bloch sphere and thereby manipulate the qubit state arbitrarily. Note, that a microwave pulse at the drive frequency mentioned above will be far detuned from the cavity resonance frequency and consequently does not constitute a measurement of the qubit state. As one can see from figure 2.3 far away from the resonator frequency the transmission amplitude will look the same irrespective of the qubit state. Therefore no information on the qubit state can be extracted from the transmission amplitude. The same holds for the phase of the transmitted or reflected signal [8], [9].

2.3. Qubit measurement protocols

Characterizing a qubit means to determine the following parameters: cavity resonance frequency ω_r , qubit resonance frequency ω_a , π -amplitude (for fixed pulse length), relaxation time T_1 , dephasing times T_2 , T_2^E , anharmonicity α and dispersive shift χ . In the following we will describe in more detail how to measure these quantities. Since the energy scales of a qubit are small, it is imperative to cool the system down to millikelvin temperatures as in all quantum experiments in order to reduce thermal noise. Otherwise thermal noise will lead to decoherence of the qubit making any controlled qubit operation impossible. Since we operate at such low temperature we can assume that the qubit will initially always be in the ground state.

2.3.1. Rabi oscillations and π -pulse

In order to obtain the quantities mentioned above, specific operations on the qubit have to be performed. One such operation that we will use frequently is a control pulse flipping the qubit state from the ground to the first excited state. On the Bloch sphere this pulse corresponds to a 180° rotation of the Bloch vector around the x-axis. For this reason the pulse is referred to as π -pulse. We also define the so called $\frac{\pi}{2}$ -pulse. On the Bloch sphere this corresponds to a 90° rotation around the x-axis which would flip the Bloch vector representing the ground state onto the equator of the Bloch sphere.

From the discussion of equation 2.7 we saw how such a rotation around the x-axis can be achieved. Furthermore, from this equation it becomes apparent that the speed of rotation is determined by the drive amplitude $\epsilon(t)$. Therefore driving the qubit with a control pulse of fixed length and variable amplitude $\epsilon(t)$ and afterwards projecting the qubit state onto the z-axis with a readout pulse will result in an oscillatory cosine pattern

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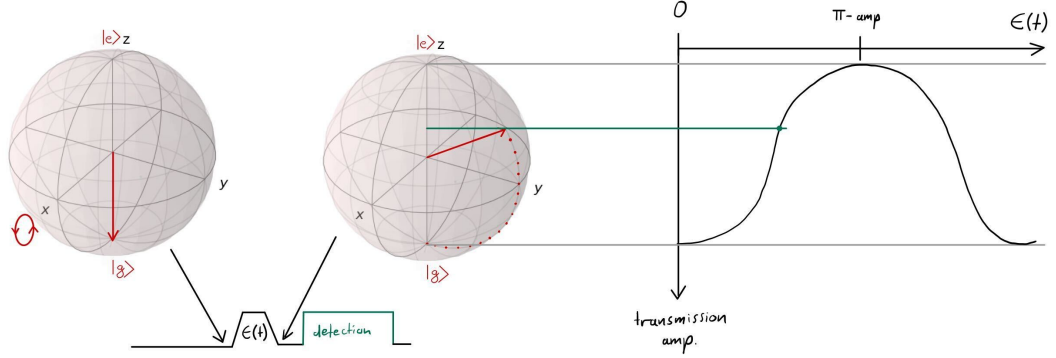


Figure 2.5.: Pulse sequence during a Rabi measurement together with the corresponding Bloch sphere representation of the qubit state. On the right-hand side the projective readout and resulting transmission signal is visualized schematically.

of the transmission amplitude of the readout pulse. This cosine pattern is called Rabi oscillations. For $\epsilon(t) = 0$ the qubit will remain in its ground state and the transmission amplitude will be at a maximum. With increasing $\epsilon(t)$ the transmission amplitude will decrease until it reaches a minimum. This happens exactly when the qubit is in the excited state. Therefore this amplitude corresponds to the amplitude necessary to perform a π -pulse. We refer to it as π -amplitude. A schematic representation of the pulse sequence during a Rabi measurement can be found in figure 2.5.

2.3.2. Cavity and qubit spectroscopy

In order to perform a Rabi measurement it is necessary to know the precise cavity and qubit resonance frequencies. The cavity resonance frequency can be determined through cavity spectroscopy in which we sweep the frequency of the readout pulse. The measured transmission amplitude will follow the behaviour depicted in figure 2.3 and from the peak we are able to determine the resonance frequency. Furthermore, if a π -pulse is performed before each readout pulse the transmission amplitude of the readout pulse will be shifted as depicted in figure 2.3. Comparing the two transmission spectra to one another allows us to determine the dispersive shift χ .

In order to determine the qubit resonance frequency we perform qubit spectroscopy where we fix the readout frequency to the cavity resonance frequency and sweep the frequency of the control pulse. Once the qubit is excited by the control pulse the transmission amplitude of the readout pulse decreases. From the resulting dip in the transmission spectrum of the readout pulse the qubit resonance frequency can be extracted.

If the amplitude of the drive is sufficiently high it is possible to excite the transmon from its $|g\rangle$ -state into its $|f\rangle$ -state via a two photon process. This will be visible as an additional dip in the transmission amplitude of the readout pulse. The separation

in frequency space of this dip and the dip caused by the qubit being in the $|e\rangle$ -state corresponds to half of the anharmonicity α .

2.3.3. Decoherence measurements

If the qubit cavity system would be completely isolated from the environment the qubit state would be stationary when pictured in a rotating frame on the Bloch sphere. Due to interactions with the environment that can never be completely suppressed, however, the system will suffer from decoherence. Following a Bloch-Redfield approach we can categorize noise into two types: transverse and longitudinal noise.

Intuitively, transverse noise will cause transitions between the ground and excited state of the qubit similar to the off-diagonal terms in an interaction Hamiltonian. This can be viewed as energy exchange between the qubit and the environment. Since superconducting qubits are usually operated at millikelvin temperatures and have transition frequencies in the gigahertz regime we assume that only de-excitations from the excited to the ground state need to be considered. In practice, however, the qubit excited state population usually is a few percent, which could be due to photons leaking into the dilution refrigerator. Neglecting this fact together with the assumption that the noise is weak and short-correlated will result in an exponential decay of the probability for the qubit to be in the excited state at time $t = \Delta t$ if it was prepared in the excited state at time $t = 0$. The lifetime T_1 associated with this exponential decay is referred to as energy or longitudinal relaxation time. It can be measured by initializing the qubit in the excited state through application of a π -pulse and then letting the qubit state evolve freely for a time delay Δt before measuring the qubit state dispersively. Since the probability of the qubit being in its ground state increases with increasing Δt this measurement will result in an exponential saturation of the transmission amplitude. From this saturation curve the lifetime T_1 can be extracted [4],[10].

Longitudinal noise will lead to fluctuations of the qubit's resonance frequency. Therefore in the rotating frame the qubit state will precess around the z-axis on the Bloch sphere according to the frequency detuning induced by longitudinal noise. Due to the stochastic nature of the noise, the qubit state will fan out into an ensemble of states. The qubit state therefore transitions from a pure state to a mixed state which can be seen as a shortening of the Bloch vector. This process is referred to as pure dephasing and results in a loss of information about the relative phase between the ground state and excited state component of the qubit state. It is one of the reasons for decoherence of a superposition state. Energy relaxation, however, also results in a loss of phase information as the excited state component of the superposition state decays and therefore destroys the information about the relative phase. The decoherence from both these phenomena is quantified by the dephasing time T_2 . It can be determined by a Ramsey measurement. The Ramsey pulse sequence and corresponding qubit state during the sequence can be found in figure 2.6. Starting from the ground state we apply a $\frac{\pi}{2}$ -pulse which flips the qubit state onto equator of the Bloch sphere and let it evolve freely for a time Δt . Here we choose, however, a drive frequency that is slightly detuned from the qubit resonance frequency, for reasons that will become apparent later. This detuning between the drive

2. Circuit quantum electrodynamics: measurements of superconducting qubits

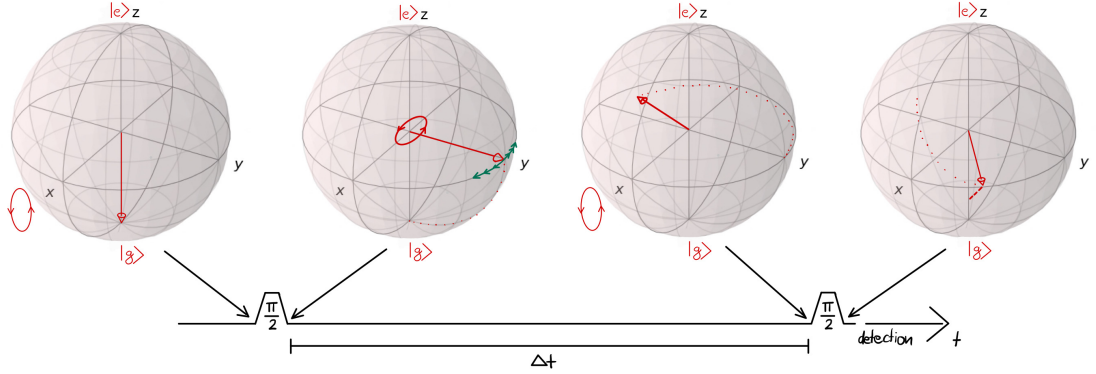


Figure 2.6.: Pulse sequence and Bloch sphere representation of the qubit state during a T_2 Ramsey measurement.

and qubit frequency leads to the Bloch vector precessing around the z-axis in a rotating frame with respect to the drive frequency. Together with decoherence shortening the Bloch vector this results in the Bloch vector spiralling inwards to the centre of the Bloch sphere. After this free evolution another $\frac{\pi}{2}$ -pulse is applied, which effectively maps this precession and fanning out of the Bloch vector onto the x-z-plane. After this second $\frac{\pi}{2}$ -pulse the readout pulse is applied which projects the Bloch vector onto the z-axis. If the time delay Δt is zero the two $\frac{\pi}{2}$ -pulses act as one π -pulse and the transmission signal should be minimal. Due to the detuning between drive and qubit frequency a cosine pattern oscillating at the frequency detuning appears. Comparing this measured detuning with the detuning that was set intentionally gives a powerful tool to accurately measure the qubit frequency. Decoherence will result in an exponential decay of the envelope of the cosine signal. This decay time is referred to as the dephasing time T_2 [4][10].

Since no energy exchange between the qubit and the environment is involved in pure dephasing this process is in principle reversible. One can make use of this fact by symmetrically adding a π -pulse between the two $\frac{\pi}{2}$ -pulses. This sequence is well known from nuclear magnetic resonance measurements as a Hahn echo sequence. Adding this π -pulse will reverse the fanning out of the Bloch vector ensemble that arises from pure dephasing and refocus the Bloch vector. The transmission signal will again follow an exponentially decaying cosine pattern. For no time delay $\Delta t = 0$, however, the Hahn echo pulse sequence consists of two consecutive π -pulses which result in the identity operation. Therefore the qubit remains in the ground state and the transition signal will be maximal. In contrast for a Ramsey sequence the transmission signal at zero time delay will be minimal. The dephasing time extracted from the exponentially decaying envelope in a Hahn echo measurement will be denoted by T_2^E [4][10].

As was alluded to above, performing a qubit characterization requires a specific order in which the respective measurements need to be performed. This will be explained further in section 4.1. Let us now focus on the concrete measurement setup that is required to perform the measurements described in this section.

Measurement setup with AWG

The setup that is used to perform the measurements described in the previous sections can be grouped into three parts: A first part to generate the microwave pulses that are used to manipulate the qubit state and for readout. Secondly the microwave cavity in which the qubit is placed and a dilution refrigerator to cool the cavity and qubit inside to millikelvin temperatures. Last but not least a part to analyse the microwave signals that are transmitted through the cavity. An overview of the microwave circuit of this setup can be found in figure 3.1. In the following we will describe the setup in more detail.

3.1. Microwave pulse generation

The core of this section is an arbitrary waveform generator (AWG) from Zurich instruments. The AWG is needed to generate the pulse shapes that will coherently drive the qubit. However, since the AWG can only output frequencies in the Megahertz range whereas typical transmon resonance frequencies lie in the low gigahertz range the AWG's output is up-converted to gigahertz frequencies using a tunable gigahertz frequency source. The up-conversion is achieved by IQ-mixing. This mixed microwave signal is combined with the microwave signal from another gigahertz frequency source, which is used to address the transmon's second excited energy state $|f\rangle$. After combining the two microwave signals the resulting signal, which will be used for manipulating the qubit state, is split in two. One part is sent to a spectrum analyzer from SignalHound to calibrate the IQ-mixer. The second part is combined with the signal from yet another gigahertz frequency source that will be used for dispersively reading out the qubit state. Since the readout pulses are long compared to the qubit drive pulses and have a rectangular pulse shape no waveform modulation is needed for these pulses. The combined qubit control and readout signal is sent to the dilution refrigerator.

The different microwave signals described above are amplitude modulated at various stages during the pulse generation as one can see in figure 3.1. Here the gigahertz frequency sources are shown in red. They are Lab Bricks from the supplier Vaunix or a SynthHD from the supplier Windfreak Technologies.

Before combining the signal that coherently drives the qubit with the signal that addresses the cavity for dispersive readout, part of the readout signal is split up and sent to the analysis part of the setup. We will come back to this reference signal in section 3.3.

3. Measurement setup with AWG

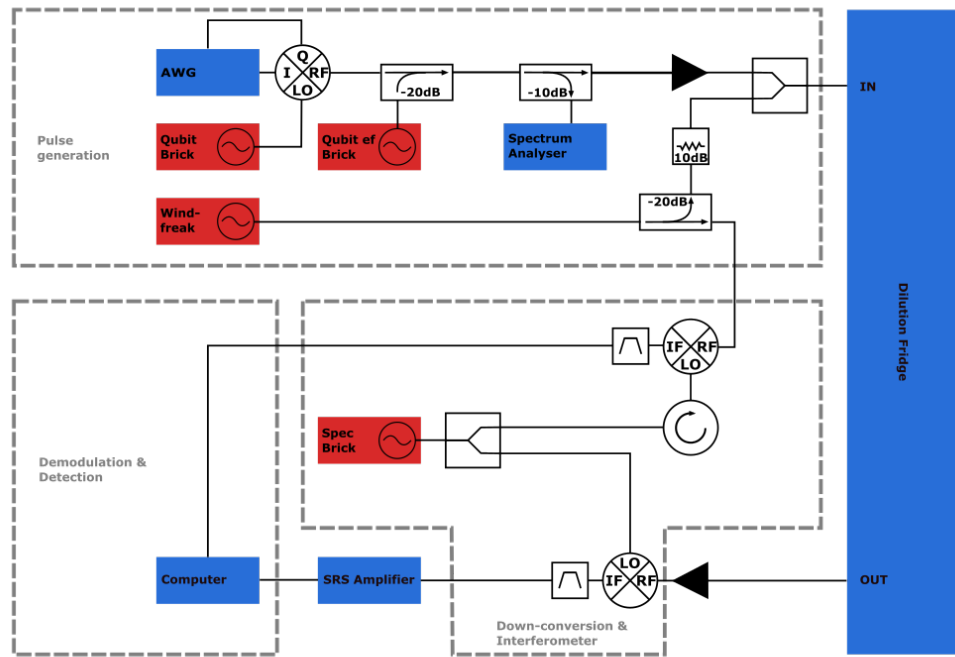


Figure 3.1.: Schematic representation of the AWG measurement setup.

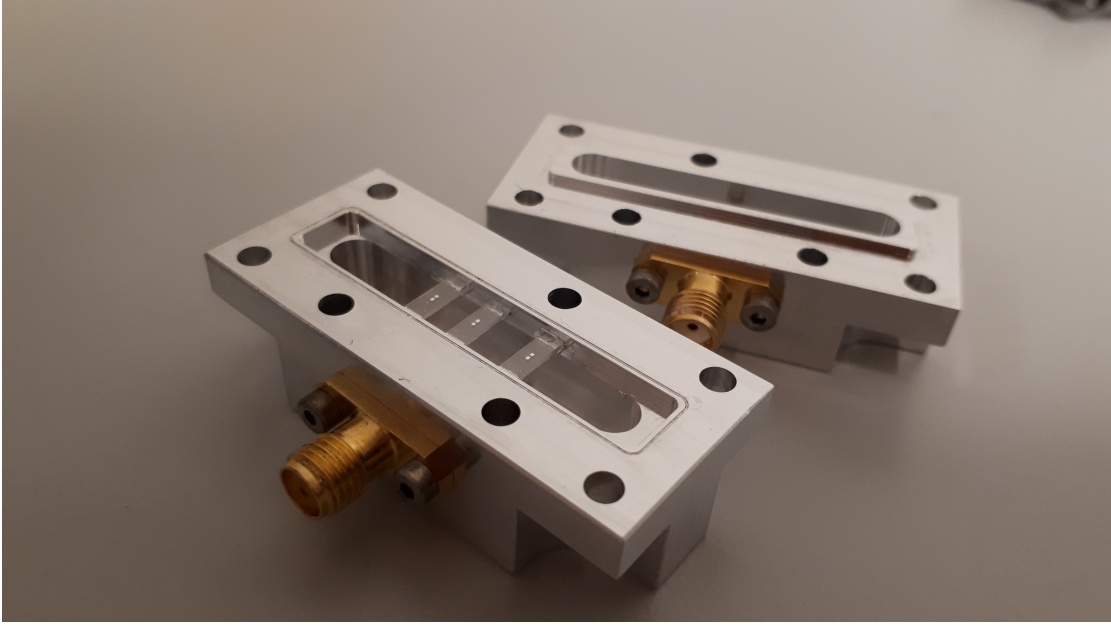


Figure 3.2.: SC transmon qubits inside an aluminium cavity.

3.2. Dilution refrigerator

The combined readout and qubit control signal is coupled into an aluminium microwave cavity. The SC qubits are placed inside of the cavity as can be seen in figure 3.2. An overview of typical parameters that our qubits have can be found in table 3.1. The aluminium cavity is mounted at the base stage of a dilution refrigerator. We use a $^3\text{He}/^4\text{He}$ dilution refrigerator from the supplier Bluefors that typically operates at a base temperature between 9 mK and 14 mK.

3.3. Signal analysis

Analyzing the signal that is transmitted through the cavity requires two steps. Firstly the signal needs to be down-converted from gigahertz frequencies to Megahertz frequencies. This is achieved by mixing the signal coming from the dilution refrigerator with the signal of a gigahertz frequency source and afterwards filtering out the high-frequency components using a bandpass filter. Secondly the down-converted signal is digitized using a waveform digitizer from the supplier AlazarTech. The digitized signal is processed and analyzed using Python.

As mentioned in the previous sections a part of the readout signal is directly sent to the analysis section. This reference signal is down-converted with the same signal that is used to down-convert the actual readout signal coming from the dilution refrigerator. The reference signal is then digitized and compared to the digitized readout signal similar as in an interferometer.

3. Measurement setup with AWG

Parameter	Measurement	Order of magnitude
Cavity resonance frequency ω_r	Cavity spectroscopy	1 GHz to 10 GHz
Qubit resonance frequency ω_a	Qubit spectroscopy	1 GHz to 10 GHz
Resistance (Josephson junction)	Resistivity measurement	10 k Ω
Π -amplitude	Rabi oscillations	-
Relaxation time T_1	-	1 μ s to 100 μ s
Dephasing time T_2, T_2^E	Ramsey, Hahn echo sequence	1 μ s to 100 μ s
Anharmonicity α	Qubit spectroscopy	100 MHz
Dispersive shift χ	Cavity spectroscopy	1 MHz to 10 MHz

Table 3.1.: Typical values of the parameters that are of interest in order to characterize a qubit.

The computer used to analyse the digitized signals is also used to control and run the experiments via a Python 2.7 script. Apart from the measurement setup described above, the experiments can also be run using a similar experimental setup that combines the functions of the different devices of the AWG setup in one platform called OPX+ that is supplied by Quantum Machines. Since the Python code controlling experiments on the AWG and the OPX+ setup are similar in their structure the goal of this project is to develop a script automating the characterization procedure that can be integrated quickly and easily on both the OPX+ and the AWG setup. For this reason the script is designed to be compatible with Python 2.7 and all newer versions.

Characterization procedure

4.1. Overview

As described previously the measurements that are necessary to fully characterize a superconducting transmon qubit have to follow a specific order. In the following section we will first give an overview over this flow of measurements and afterwards focus in more detail on the individual steps during this procedure together with the analysis necessary to extract the quantities of interest.

Before the characterization procedure is started, typically two steps are performed at room temperature in preparation: Firstly, we measure the resistivity across the transmon's Josephson tunneling junction, from which an estimate of the qubit resonance frequency can be made. Secondly, after mounting the qubits inside of the cavity the cavity's resonance frequency at room temperature can be determined via reflection or transmission with a vector network analyser (VNA). These quantities will be the starting point of the characterization procedure. After the measurements the cavity together with typically three qubits inside is mounted into the dilution refrigerator and cooled down to 80 mK. At this point the characterization begins. A schematic overview of the following steps can be found in figure 4.1. First we find the cavity resonance at 80 mK, which is necessary in order to tune the readout pulse to the cavity resonance frequency to perform dispersive readout of the qubit state. Then the readout pulse's power is optimized to get the best possible signal-to-noise ratio. Next we search for qubit resonance candidates and test their response to changes in the control pulse's power, which we refer to as the probe power. This gives a first indication of whether the resonance dip belongs to a qubit or not. Depending on the outcome of this measurement we either discard the candidate or continue with a Rabi measurement, which will give us a clear indication of whether the resonance corresponds to a qubit or not and if so it also gives the π -amplitude. By performing a Ramsey measurement and afterwards repeating the Rabi measurement we can calibrate the π -pulse to the qubit. Now all other quantities of interest can be extracted in arbitrary order: anharmonicity α , relaxation time T_1 , dephasing times T_2, T_2^E and dispersive shift χ by performing the measurements described in section 2.3. In the following we discuss the concrete implementation of this procedure.

From the overview above it becomes apparent that the challenge in the automation of the characterization procedure lies in the first steps up to and including the calibration of the π -pulse. The success of the characterization algorithm and quality of the obtained quantities depends on these first steps. Making sure that this part is performed robustly

4. Characterization procedure

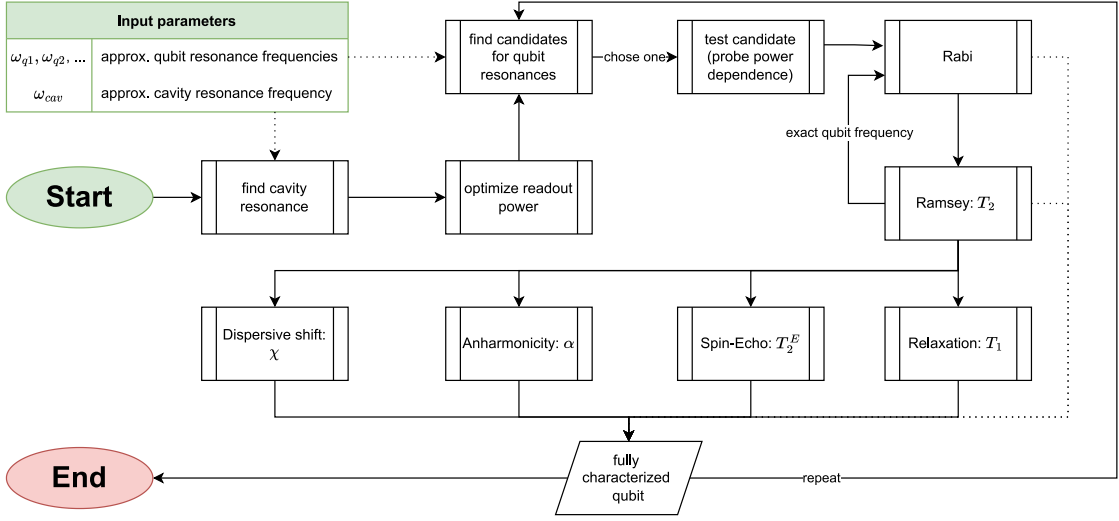


Figure 4.1.: Flow diagram showing the general procedure for characterizing a qubit.

and accurately is therefore imperative in order to ensure good and helpful results.

4.2. Analysis Tools

We first introduce the analysis tools that are used throughout the characterization procedure. From the previous section it has become clear that at every point of the procedure data needs to be analysed in order to use the results as input for the next measurements. Therefore it is crucial to make sure that the analysis works robustly and reliably. In order to ensure this we use two tools: A peak finding algorithm that is used to get initial estimates of the fitting parameters and the coefficient of determination in order to assess the goodness of fit.

4.2.1. Peak detection

The peak detection algorithm that we use is taken from reference [11]. In order to fine tune the algorithm the noise level and minimum distance between neighbouring peaks are required as inputs. Details on how these are estimated can be found in section A. The peak finding algorithm returns a list of dips and list of peaks in the data which are used for two purposes: Firstly to detect peaks and dips in qubit and cavity spectroscopy data and thus to find qubit candidates. Secondly we use it to estimate the initial values of the fitting parameters that are given to the fitting function `scipy.optimize.curve_fit()`. Good starting values are essential such that the function converges and fits the correct features. For Lorentzian lineshapes the output is used to estimate the peak height and center frequency. For oscillatory data we use it to find the minima and maxima of the oscillations and from that extract the oscillation period. Furthermore, we are able to extract the exponential decay of the signal in T_2 and T_2^E measurements from the decaying

height of the peaks and dips. Combining the peak and dip data with the average of all data points and the minima and maxima of the data allows us to extract all other initial parameters such as the offset or amplitude of the fit function.

The peak finding algorithm performs well for most data sets. For broad qubit spectroscopy sweeps spanning several hundreds of megahertz, however, many false positive peaks are detected. In order to remove the false positives in this case we combine the peak detection algorithm with a R^2 -test that will be explained in the next section.

4.2.2. Coefficient of determination

Consider a data set of N data points where the i -th data point is given by (x_i, y_i) . The coefficient of determination is defined as

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{\sum_{i=0}^N (y_i - f_i)^2}{\sum_{i=0}^N (y_i - \bar{y})^2} \quad (4.1)$$

where $\bar{y}$ is the average y-value of all data points and f_i is the prediction of the fit at x_i . A visual representation of the meaning behind the different terms in equation 4.1 can be found in figure 4.2. The sum in the numerator of the second term SS_{res} is proportional to the residual sum of squares of the fit which is visualized by the blue squares in figure 4.2. Intuitively, it quantifies the error of the fit. If the fit represents the data well this term will be small. If the fit does not describe the data well this term is large. In order to convert this value to a meaningful scale that is independent of the number of data points, we divide by the sum in the denominator of the second term SS_{tot} which is proportional to the variance of the data. This sum is visualized by the red squares in figure 4.2. For a good fit SS_{tot} will be much larger than SS_{res} and therefore R^2 will be close to one. If the fit does not model the data well SS_{tot} and SS_{res} are of similar magnitude which results in R^2 being close to zero. Since both SS_{res} and SS_{tot} are positive, the coefficient of determination is always smaller than one and typically larger than zero [12].

It is clear that a stronger signal compared to the noise level will increase the variance of the data points and therefore increase SS_{tot} while SS_{res} remains unchanged. As can be seen from equation 4.1 an increase in signal-to-noise ratio will therefore increase R^2 . We conclude that therefore high R^2 values are desirable in our case.

In regression analysis R^2 is used as a measure for the goodness of fit. Although no value exists that can define what constitutes a decisive R^2 -value we can use R^2 as a tool to differentiate between random noise and features such as Lorentzian lineshapes or Rabi oscillations [13]. For these R^2 -tests we determined threshold values for the coefficient of determination as described in section B. If R^2 lies above the threshold value we assume that the feature that we expect such as a Lorentzian lineshape for example is present in the data. If R^2 lies below the threshold we assume that the feature we expect is not present in the data.

Combining the peak detection algorithm with an R^2 -test allows us to remove false positive dips when analysing broad qubit spectroscopy sweeps. For each detected dip we take a frequency window of 10 MHz centered around it from the data. This sub-set of

4. Characterization procedure

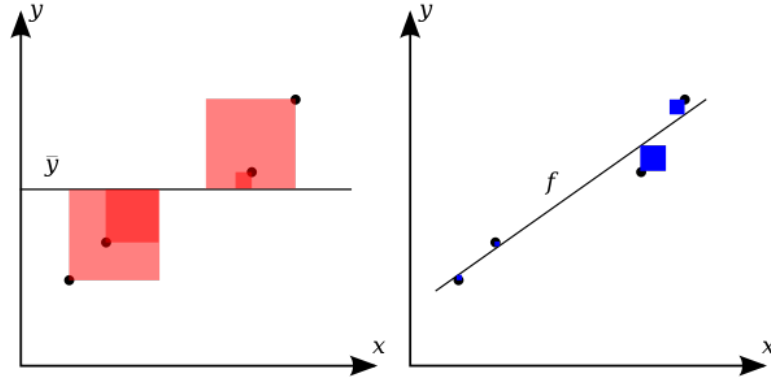


Figure 4.2.: Visualization of the different terms in equation 4.1 in the case of a linear regression. The total area of the red squares on the left-hand side corresponds to SS_{tot} . The total area of the blue squares on the right-hand side corresponds to SS_{res} .

the data is fitted with with a Lorentzian and a R^2 -test is performed. If the dip fails the R^2 -test it is removed eliminating false positive peaks that cannot be distinguished from noise.

4.2.3. Savitzky-Golay-filter

We are successfully able to determine dips in the transmission spectra with the procedure described above. However, so far we haven't utilized the fact that dips corresponding to $|g\rangle \leftrightarrow |f\rangle$ two photon processes are much narrower than dips corresponding to $|g\rangle \leftrightarrow |e\rangle$ transitions at the same probe power, as can be seen from the data shown in dark blue in figure 4.3a. In order to filter the narrow peaks out during qubit detection we use Python's `scipy.signal.savgol_filter` function, which applies a Savitzky-Golay-filter to the data. The idea behind a Savitzky-Golay filter is to split up the data into successive sub-sets of a specific length, in our case roughly 30 MHz. For each sub-set or window the data is fitted with a 4th order polynomial by the method of least squares. From this fit an expected value at the center of the window can be calculated. This value will give the value of the filtered signal. The window then moves across the data by taking up one point on the right and dropping one on the left of the window and repeating the procedure. In order to speed up the computation the filter performs a discrete convolution instead of least squares fitting [14]. An example of how this filter acts on data can be found in figure 4.3a in light blue.

Now that we have introduced all analysis tools that are used in the characterization we focus on the details of the procedure.

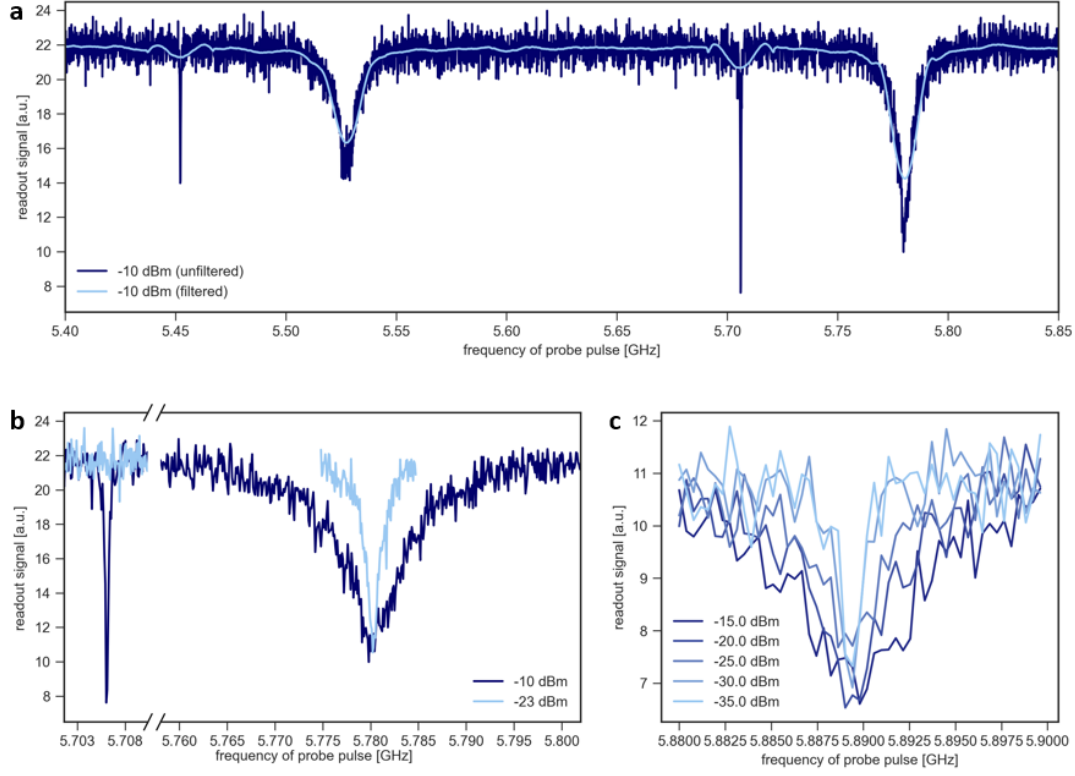


Figure 4.3.: (a) Qubit spectroscopy data for two qubits inside a microwave cavity in dark blue. The two wide dips correspond to $|g\rangle \leftrightarrow |e\rangle$ transitions, whereas the two narrow dips correspond to $|g\rangle \leftrightarrow |f\rangle$ two photon processes. In light blue the data is shown after applying a Savitzky-Golay-filter. (b) Zoom-in on the data shown in (a) together with additional data of the same qubit taken at lower probe power. (c) Qubit spectroscopy data for the $|g\rangle \leftrightarrow |e\rangle$ transition taken at different probe powers, demonstrating the power dependence of the transition linewidth.

4. Characterization procedure

4.3. Cavity resonance and readout power optimization

The first step of the characterization procedure requires calibrating the readout pulse. A flow diagram of this step can be found in figure 4.4. As described above we first have to find the cavity resonance frequency. For this we start with an intermediate power of the readout pulse. This power only serves as a starting point for the procedure, it should, however, not be chosen too high or too low. If it is chosen too low the cavity resonance can not be distinguished from noise which will increase the duration of the characterization procedure. If it is chosen too high the cavity resonance becomes non-linear due to interactions between the cavity and qubit and it becomes more difficult to correctly determine the cavity resonance frequency. Furthermore, long sweeps at high power can lead to heating of the base stage due to dissipation. The exact value of the intermediate readout power can be found in section C together with the values of all other standard parameters.

From experience we know that the cavity resonance frequency increases by approximately 80 MHz when the cavity is cooled from room temperature down to 80 mK. Therefore we perform cavity spectroscopy from 50 MHz below up to 250 MHz above the cavity resonance at room temperature. We analyse the transmission spectrum with the tools described in section 4.2 and perform a R^2 -test. If the data passes the R^2 -test we conclude that we have found a cavity and continue with optimizing the readout pulse's power. If it fails we repeat the cavity spectroscopy. For each repetition we increase the power of the readout pulse and choose the largest possible range of resonance frequencies that can be expected for our cavities. This largest possible frequency range is also chosen for the first cavity spectroscopy sweep if the cavity resonance at room temperature is unknown. Since such a broad sweep takes about one hour this should be avoided by always stating the cavity resonance at room temperature. If the cavity resonance cannot be found after 4 repetitions the code terminates.

Having determined the cavity resonance frequency we continue with optimizing the readout power to find a more suitable value than our starting value. For this we perform cavity spectroscopy from 10 MHz below up to 10 MHz above the cavity resonance at several different readout powers. An example of such cavity spectroscopy measurements can be found in figure 4.5a. We then calculate the coefficient of determination for each trace and plot the coefficient of determination versus power as seen in 4.5b. At very low readout powers the resonance cannot be distinguished from the noise floor which will result in R^2 -values close to zero. With increasing readout power the signal-to-noise ratio improves which will lead to an increase in the R^2 -value. At high readout powers, however, the cavity's response will become non-linear and the transmission spectrum will deviate from the expected Lorentzian lineshape. Therefore the R^2 -value of a Lorentzian fit of the data will decrease. This will give a maximum of the R^2 -value at intermediate powers. The maximum is determined by fitting the R^2 versus readout power trace with a parabola. We expect that the signal-to-noise ratio is relatively high but non-linear effects are still small at this maximum. We now repeat the cavity spectroscopy with this new readout power. From this last measurement we extract the exact cavity resonance frequency,

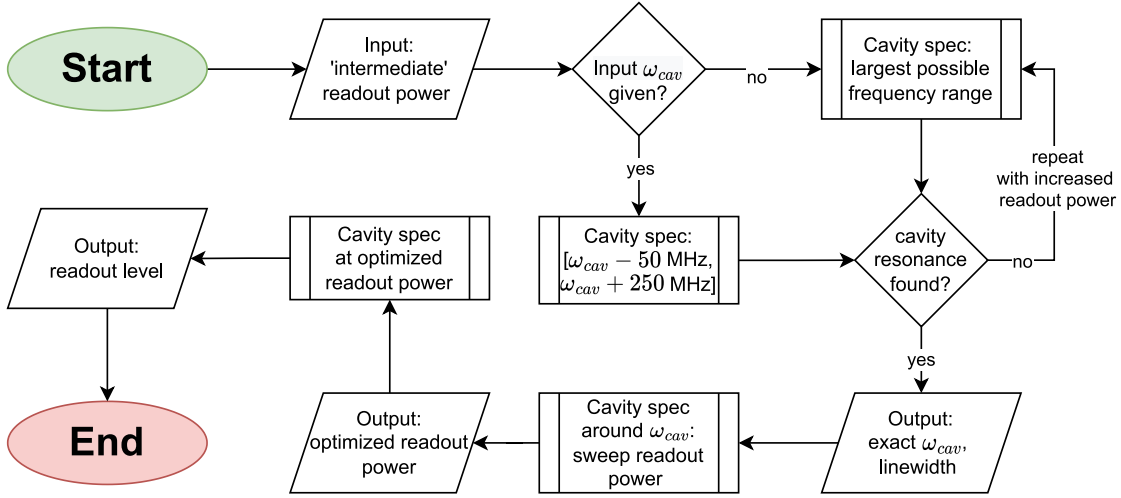


Figure 4.4.: Flow diagram showing how the cavity resonance frequency is found and how the readout power is optimized.

linewidth and the amplitude of the transmission spectrum which we call readout level. Information about the readout level is not necessary for the characterization procedure but could be used in sanity checks during later stages of the characterization procedure. Currently this is not implemented.

Having found the exact cavity resonance frequency and optimized the power of the readout pulse concludes the calibration of the readout pulse. The next step of the characterization procedure is to find and test qubit resonances.

4.4. Testing qubit candidates

The results of the resistivity measurements that were performed in preparation for the qubit characterization procedure will serve as a rough estimate of the qubit resonance frequencies. From experience we know that the actual resonance frequencies might deviate by hundreds of megahertz. For this reason we decided to perform one qubit spectroscopy measurement, where we sweep a broad frequency range for the control pulse. From this single measurement we extract all qubit candidates at once instead of performing multiple sweeps close to the estimated qubit resonances. Although this option is not the most time efficient one, it appears to be the most reliable and robust solution. Furthermore, this approach makes it significantly easier to retrace and understand the algorithm's decisions.

A flow diagram of the search for and testing of qubit candidates, which will be discussed in more detail in the following, can be found in figure 4.6. The control pulse's power for the broad sweep that is used to find candidates for qubit resonances is chosen to be an intermediate power. This power does not need to be well calibrated, it should, however, on the one hand not be too high as this will lead to heating of the base stage

4. Characterization procedure

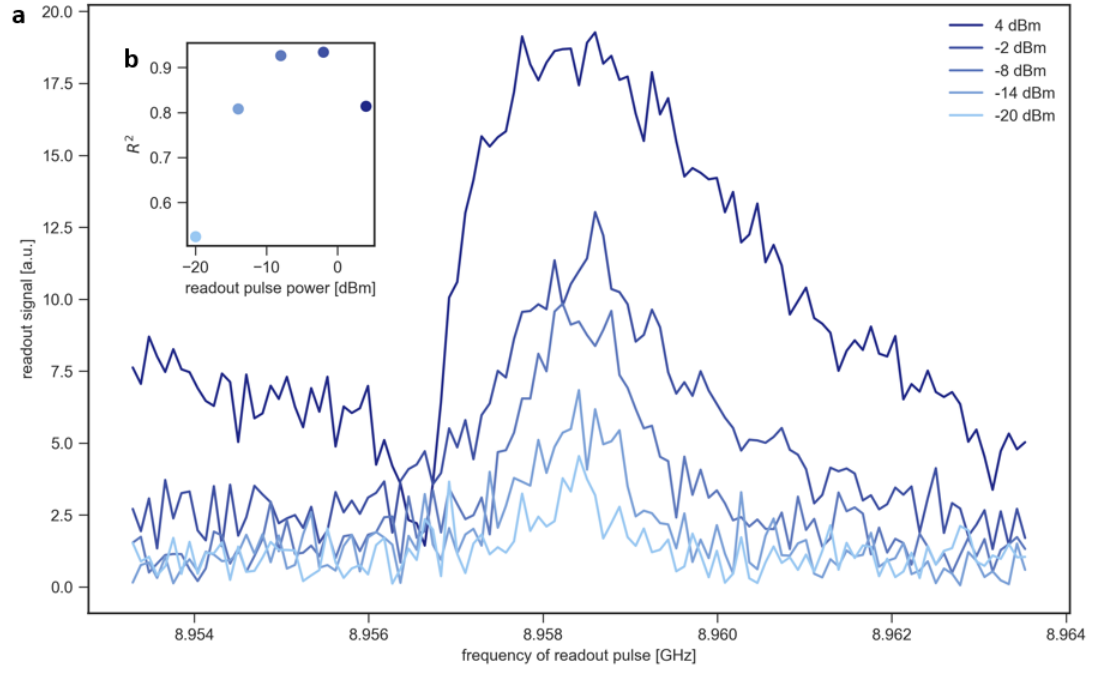


Figure 4.5.: (a) Cavity spectroscopy data of microwave cavity at different readout powers. (b) Coefficient of determination R^2 calculated for a Lorentzian fit of different traces shown in (a). Each point corresponds to one cavity spectroscopy measurement at a fixed readout power.

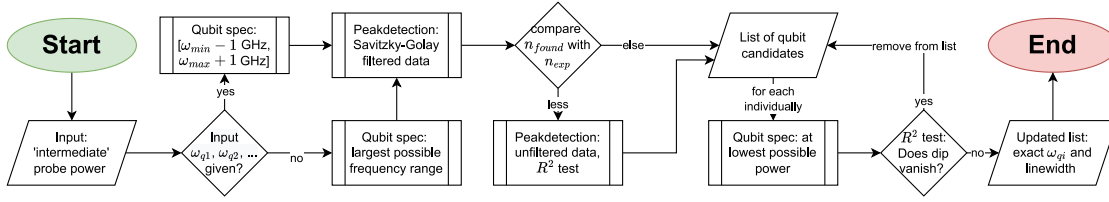


Figure 4.6.: Flow diagram showing how qubit resonances are found screened for a first time.

of the dilution refrigerator. Furthermore, at high powers more dips appear in the qubit spectroscopy measurements that correspond to processes which are not related to the $|g\rangle \rightarrow |e\rangle$ transition and therefore not of interest, as seen in figure 4.3a,b. On the other hand the probe power should not be too low as the linewidths of the qubit resonances will decrease with decreasing power, as shown in figure 4.3c. Thus a smaller stepsize would be necessary for the qubit spectroscopy sweep, which would therefore take longer to perform. The frequency range over which we sweep the control pulse during the broad qubit spectroscopy depends on whether estimates of the qubit resonances are given or not. If estimates are given, we measure from one gigahertz below the lowest estimated qubit frequency up to one gigahertz above the highest estimated qubit frequency. If no estimate is given, we choose the largest possible frequency range that qubit resonances can be expected to lie in. In our case this range is 4 GHz to 8 GHz and one sweep over this range will take roughly 12 hours, which will significantly increase the duration of the characterization procedure. This should therefore be avoided.

In order to locate qubit resonance candidates we first analyse the qubit spectroscopy data after post processing it using a Savitzky-Golay filter. More details on this filter and the peak detection analysis are described in section 4.2. This gives us a first list of potential qubit resonances. We compare the number of qubit resonance candidates on this list with the number of qubits that are inside of the cavity. If more candidates than expected or equally many candidates as expected were found we proceed with the next step of testing the candidates. If less candidates were found than expected, we analyse the raw, unfiltered data with a combination of the peak detection algorithm and the R^2 -test, as described in section 4.2.1. This gives us a new list of qubit resonance frequency candidates with which we proceed to the next step of testing the candidates.

As a first test we examine the resonance candidate's behaviour at low powers of the qubit control pulse. If the dip in transmission amplitude corresponds to a two-photon process for example it should vanish at low power, as seen in figure 4.3b. We perform qubit spectroscopy at the lowest possible probe power that can be set by the instruments, where we sweep the control pulse's frequency in a range of 10 MHz centered around the candidate resonance frequency. In order to determine whether the dip vanished or not we perform a R^2 -analysis. If R^2 lies below the threshold we discard of the candidate. Otherwise it remains in the list. In theory the dip should also vanish for qubits at very low powers. In practice, however, such low powers are not realized by the measurement setup. The dip is therefore always present for qubits even at the lowest possible power,

4. Characterization procedure

as can be seen in figure 4.3c.

This first test gives a good indication of whether the dip corresponds to a qubit or not and also gives a more precise value of the resonance frequency. With the updated list we proceed with the next step of the procedure in which we test the candidates more thoroughly and calibrate the π -pulse for each qubit.

4.5. Coherent drive and characterization of qubit

Since Rabi oscillations can be observed for any coherent two-level system we can use this fact to further test the qubit candidates that we got from the procedure up to this point. The procedure described in the following has to be repeated for each qubit candidate. A flow diagram of the process can be found in 4.7.

In order to test whether the candidate displays Rabi oscillations or not we perform a series of Rabi measurements. From the previous section we have the candidate resonance frequency that we use as the drive frequency for the control pulse. Since we sweep the control pulse's power in the Rabi measurement we have to choose a fixed pulse length. For the first Rabi measurement we start with a control pulse length of 10 ns. In order to decide whether Rabi oscillations are visible in the data we perform an R^2 -test. If the R^2 -value lies above the threshold, the system shows Rabi oscillations which is a strong indication of it being a qubit. Therefore we can continue with calibrating the π -pulse. If the R^2 -value lies below the threshold we change the control pulse's length and perform a new round of Rabi measurements and R^2 -tests. During the n -th round of repetitions we perform two Rabi-measurements, one with a pulse length of $2^n \times 10$ ns and one with a pulse length of $2^{-n} \times 10$ ns. If no oscillations are detected after 6 rounds of Rabi measurements, we conclude that the candidate does not correspond to a coherent TLS and thus discard it and proceed with the next candidate.

In the case of Rabi oscillations, we move on to the next step where we determine the π -amplitude from the last set of data. In order for our instruments to be able to perform a π - and $\frac{\pi}{2}$ -pulse, we want the π -amplitude to lie between 0.1 and 0.75. These values are given in normalized power with respect to the instruments maximum power output. If the π -amplitude is lower than 0.1 we decrease the length of the control pulse by 50%. If the π -amplitude is higher than 0.75 we increase the length of the control pulse by 90%. Then we perform another Rabi measurement and R^2 -test and compare the resulting π -amplitude with our desired range. This process is repeated until either the π -amplitude lies in the desired range or the data fails the R^2 -test. If it fails the R^2 -test the candidate is discarded and we move on to the next candidate. If the π -amplitude lies between 0.1 and 0.75 we move on to the next step, which consists of fine tuning the π -pulse.

In order to fine tune the π -pulse we perform a Ramsey measurement where we set an artificial detuning δ_{set} of 500 kHz between the qubit's resonance frequency and the control pulse. As described in section 2.3.3 we can determine the actual detuning δ_{real} from the measurement results and compare it to δ_{set} . Ideally δ_{set} would be equal to δ_{real} . This will, however, usually not be the case. If the difference between the two is larger than 10% of δ_{set} we correct our value of the qubit's resonance frequency by

4.5. Coherent drive and characterization of qubit

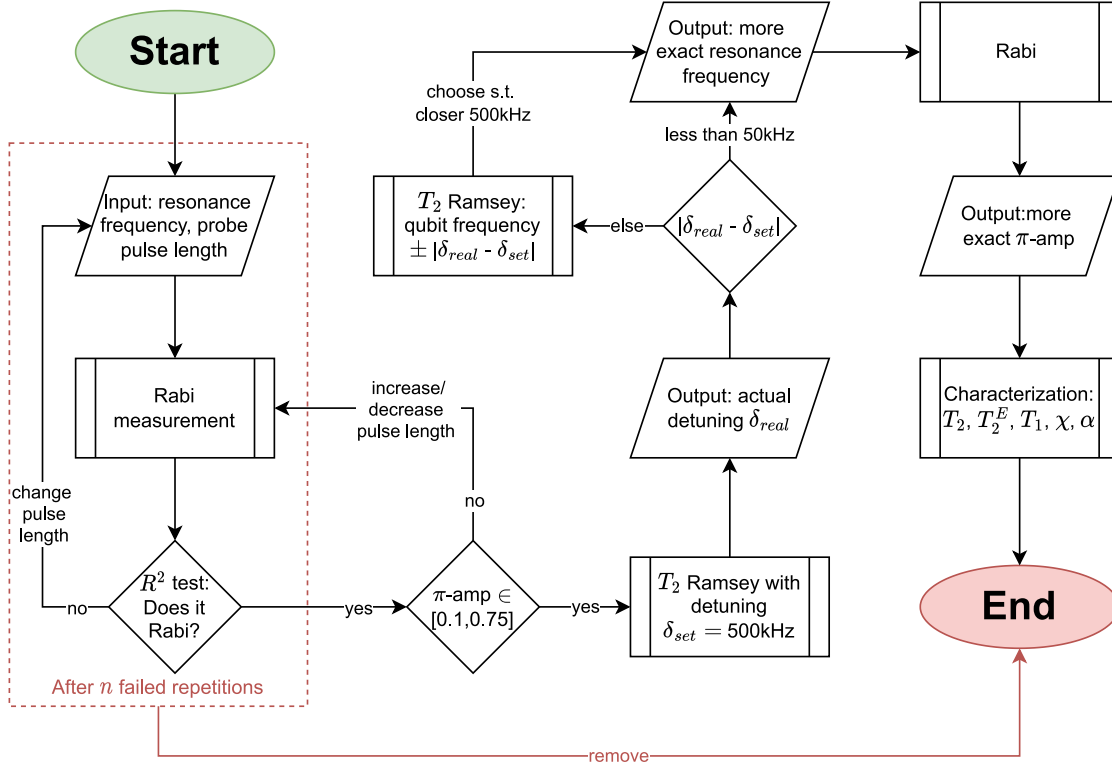


Figure 4.7.: Flow diagram showing how the previously found qubit candidates are tested, the coherent drive pulse is calibrated and how the qubit is then characterized.

$|\delta_{set} - \delta_{real}|$. Since we cannot immediately tell whether our estimate of the qubit resonance frequency should be increased or decreased, we first increase it and perform another Ramsey measurement. From this measurement we get a new value of the detuning δ_{real}^{up} . Then we repeat this process for a decreased estimate of the qubit resonance frequency, from which we determine δ_{real}^{down} . We compare $|\delta_{set} - \delta_{real}^{up}|$ to $|\delta_{set} - \delta_{real}^{down}|$ and choose the new estimate of the qubit resonance frequency based on which estimate gives the smaller difference between set and measured detuning. We repeat the Rabi measurement with this new, very precise estimate of the qubit resonance frequency and extract the π -amplitude. This concludes the calibration of the π -pulse.

All other measurements that are necessary to characterize the qubit can now be performed. Namely T_1 , T_2^E , T_2 (repeated), χ , α are measured. After all quantities of interest are determined we move on to the next qubit candidate. Once the last candidate has been examined the characterization procedure terminates.

Results

The Python script that performs the characterization procedure can be grouped into three layers. The first layer, the presentation layer, is responsible for the user experience. It manages the interaction between the user and the software. The second layer, the business layer, determines how the automation of the characterization procedure is realized. The logic of the procedure and analysis involved in it has been discussed in the previous section. The third and lowest layer manages the communication between the measurement setup and the business layer. It translates the commands of the business layer into commands that can be processed by the measurement setup.

For the presentation layer we developed a general user interface (GUI) based on the Python tkinter package. The GUI allows you to pass all the relevant information such as the estimated cavity and qubit frequencies to the software. Via the GUI you can choose which measurements to perform as well as the name of the output files that will be generated. Furthermore, the GUI gives an overview over all parameters listed in section C together with a short description of each parameter's function and the possibility to change the parameter values. A screenshot of the GUI can be found in figure 5.1. The GUI uses Python's threading module. Since this module might not be compatible with the measurement setup's code the use of different threads can be turned on or off directly in the *characterization_GUI.py* file via the value of *use_threading*.

More details on the business layer can be found in section 4 or the *characterization.py* and *analysis.py* files.

The base layer setting up the communication between measurement apparatus and software is established in the *AWG_measurements.py* file. This is the only part of the software that is specific to the setup. Therefore, this is the only layer that needs to be adapted when implementing the characterization software on a new setup like the OPX+ setup. Since the software on the AWG and OPX+ setup are very similar this can be achieved quickly and easily.

In order to start the characterization software the *run_characterization_procedure.py* file needs to be run. There are three different modes available to perform a characterization. Whether the GUI is turned on or off is determined by the value of *run_with_GUI*. The GUI-debugging mode can be turned on or off via *GUI_debugging_mode*. If the debugging mode is turned on, the buttons starting the characterization procedure and making the report are replaced by dummy buttons. Furthermore, a testing mode for the characterization procedure exists. If *characterization_dummy_data* is *True* the characterization algorithm is fed data from previous cooldowns instead of performing the actual measurements.

5. Results

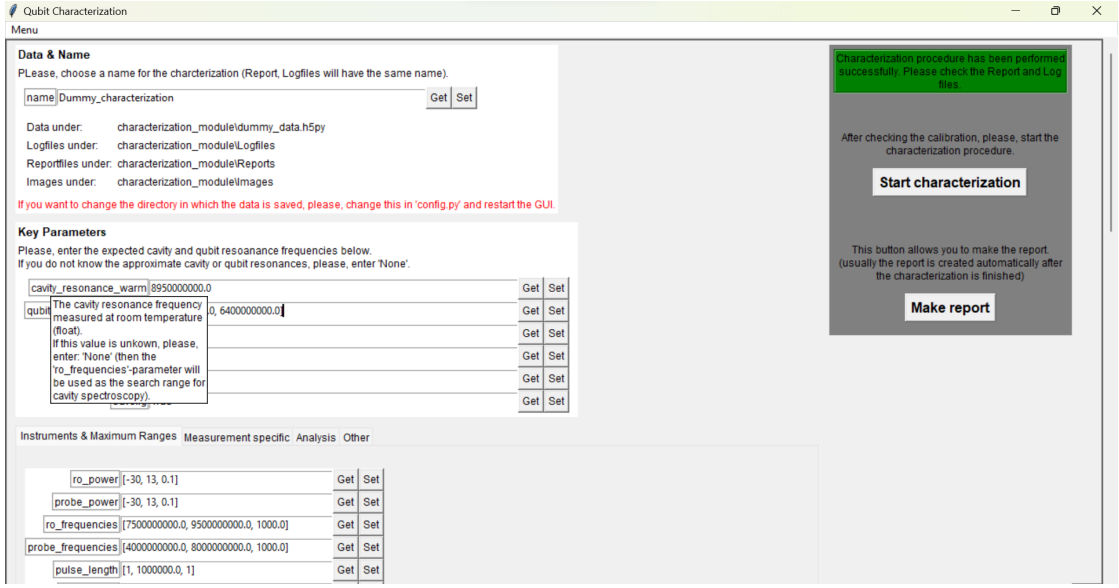


Figure 5.1.: Screenshot of the GUI. In the top right corner the buttons for starting the characterization procedure and making the report can be found.

After the characterization procedure is finished, two outputs are automatically generated: a log file and a PDF report. The log file documents the actions the algorithm took. This includes the measurements that were performed together with the settings used, the analysis results and the decisions based on this information. The log file makes it possible to easily retrace and understand the algorithms actions similar to a lab book. An example of how such a log file can look like can be found in section D. The PDF report summaries the results of the characterization procedure. Here the most important measurements and the quantities of interest that were extracted from these are displayed. The report can be exported into the OneNote lab notebook. An example of how such a report might look like can be found in section E.

A previous version of the software with reduced functionality was tested on 5 qubits that were each bonded to a HBAR. In this test all 5 qubits could be detected and characterized successfully.

Limitations of the characterization code and outlook

In some situations the software might not be able to characterize a qubit successfully due to inherent limitations. We want to discuss such limitations in the following.

As is apparent from the previous sections some parameters are not changed and optimized throughout the characterization procedure. Among these are the length of the readout pulse the sampling length of the waveform digitizer or the number of measurements *naverage* that are averaged to obtain one data point.

The readout pulse length and sampling length need to be chosen with regard to the quality factor of the cavity. Since the quality factor of the cavity is usually tuned to lie in a similar regime for all cooldowns no change in the readout and sampling length should be necessary. In case the cavity has an exceptionally low or high quality factor tuning the readout and sampling length should be done manually.

The number of measurements *naverage* influences mainly the signal-to-noise ratio and the duration of a measurement. For the detection of qubits a very good signal-to-noise ratio is not critical. Therefore a standard value of *naverage* = 3000 should be sufficient in most cases. If the out-coupling rate of the cavity is, however, very low a higher *naverage* might be required. The exact values of these three parameters discussed above are very difficult to optimize automatically as they need to be chosen from experience.

Furthermore, due to the length of broad qubit spectroscopy sweeps we decided to only perform one broad sweep to find qubit candidates. If the candidates cannot be detected at this point they will not be characterized.

Other situations that could pose a challenge to the software are situations in which the qubit behaves unusually deviating from the behaviour discussed in the theory section of this report. These could be situations in which two qubit resonance frequencies lie close to one another such that their coupling is not negligible anymore or if the qubit and cavity resonance frequencies lie very close to one another such that the system does not operate in the dispersive limit. These situations could lead to a different functional dependence of the measurement results. Differentiating between these different cases cannot be achieved with the coefficient of determination and would require different analysis tools. The approach that we chose can only be used to tell apart random noise and one type of behaviour that can be expected from a qubit.

Instead of using a peak finding algorithm and the coefficient of determination for finding qubit resonances and making decisions other more advanced tools could be used

6. Limitations of the characterization code and outlook

in the future. Neural networks for example might perform better than our approach at identifying qubit resonances and at answering questions like whether Rabi oscillations can be observed or not. The data necessary to train a neural network for such tasks could be taken from old cooldowns. Furthermore, neural networks could be useful in complex situations such as the case of two qubits that are close together in their resonance frequencies.

Apart from this parameters like the readout level that are currently not considered could be included in the analysis. The readout level extracted from cavity spectroscopy could for example be used to check the strength of the signal during qubit spectroscopy. Adding corresponding warning messages if for example the signal is lower than expected might be beneficial to the user experience.

Conclusion

Using the coefficient of determination, a peak finding algorithm and a Savitzky-Golay filter we developed tools in order to automatically analyse the measurements performed during the characterization of a superconducting qubit which is coupled dispersively to a microwave cavity and make decisions based on the analysis results. With our approach we successfully automated the characterization of superconducting qubits via a Python script that is compatible with Python 2.7 and all newer versions. Automating this process promises to free up a lot of time that can in the future for example be spend on the design and fabrication of the transmon qubits. Furthermore, the characterization can be run overnight making optimal use of the measurement infrastructure.

After first tests of the code that yielded promising results the next step will be to use the code in everyday lab work and thus test it on a variety of different experiments. Based on the software's performance improvements can be made in the future or new approaches like the use of neural networks in the analysis of measurement data can be explored.

References

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Fine-tuning the peak detection algorithm

The peak detection algorithm requires the noise level and minimum separation between peaks as inputs. In the following we discuss how these were determined.

For each different type of measurement we define a minimum length scale of interest as can be found in A.1. We assume that all oscillations on length scales lower than this scale are due to noise and that the signal should therefore be approximately constant over this length scale. We estimate the level of noise by binning the data where each bin's length is given by the minimum length scale. For each bin we calculate the average of the data points and from that extract the variance of the data inside the bin. The noise level is extracted from this by taking the square root of the average of the variances over all bins. This procedure corresponds to approximating the data by a step function where the length of each step is the minimum length scale of interest. The noise level then corresponds to the sum of squares of the distance between the data and the step function.

The minimum separation of peaks depends on the type of measurement that is analysed. For oscillatory data the minimum distance between two peaks is taken to be five times the minimum length scale. For all measurements in which we expect only one peak or dip such as cavity spectroscopy the minimum separation is taken as one twentieth of the window length of the data that is analyzed. For qubit spectroscopy the minimum separation is taken as 30 MHz and for alpha measurements the minimum separation is taken as 10 MHz.

minimum length scale for ...	value
cavity spectroscopy (including χ)	100 kHz
qubit spectroscopy (including α)	100 kHz
Rabi measurements	0.04
T_1, T_2, T_2^E measurements	100 ns
Ramsey measurements (to determine qubit frequency)	20 ns

Table A.1.: Minimum length scales of interest together with their values.

Determining threshold values for the coefficient of determination

As described in section 4.2.2 we use thresholds for the coefficient of determination in order to determine whether we see features such as Lorentzian dips/peaks or Rabi oscillations for example in the data. An overview of all thresholds can be found in table B.1. In the following we will describe how we determined the threshold values that we use.

In the case of Lorentzian peaks or dips we measured the transmission spectrum of two different cavities at various readout powers close to their resonance frequency. The data (black) can be seen in tables B.2 and B.3 together with a Lorentzian fit (red) of the data and the corresponding R^2 -value. For R^2 -values below 0.383 it is difficult to decide whether a peak is present in the data or whether this is only noise. For R^2 -values above 0.549, however, a peak is clearly visible. For this reason we decided on 0.4 as the threshold value in case we are examining a cavity spectroscopy measurement and 0.5 as the threshold value in case we are examining a qubit spectroscopy measurement.

In the case of Rabi oscillations we performed Rabi measurements at various detunings between the qubit and the control pulse. The data (black) can be seen in table B.4 together with a Cosine fit (red) of the data and the corresponding R^2 -value. As one can see the data clearly looks like an oscillation for all R^2 -values above 0.219. Therefore we set the threshold at 0.15.

In the case of T_1 , T_2 or T_2^E measurements the coefficient of determination is not used for any decision making purposes and only influences the warning messages at the beginning of the report. For this reason no tests were done for these types of measurements and the threshold was set to a value of 0.45.

Data type	R^2 threshold value
Lorentzian peak (cavity)	0.4
Lorentzian dip (qubit)	0.5
Rabi oscillations	0.15
T_1 , T_2 , T_2^E traces	0.45

Table B.1.: Threshold values for the different types of data.

B. Determining threshold values for the coefficient of determination

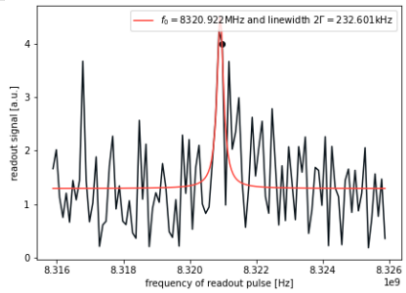
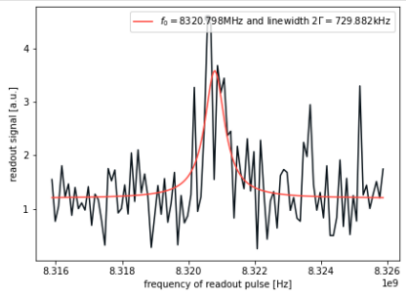
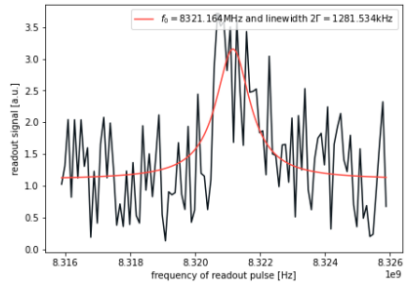
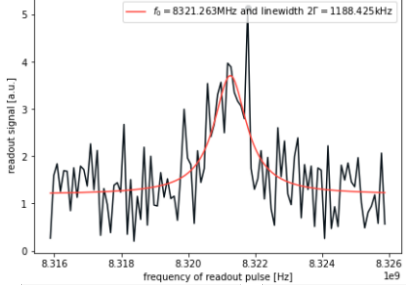
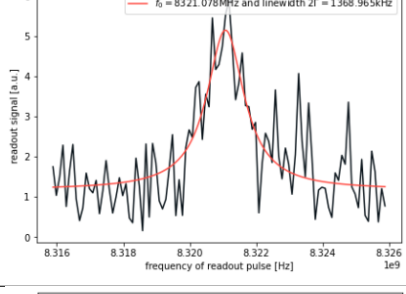
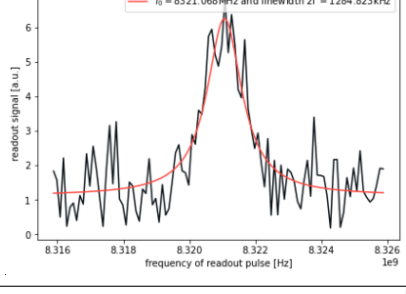
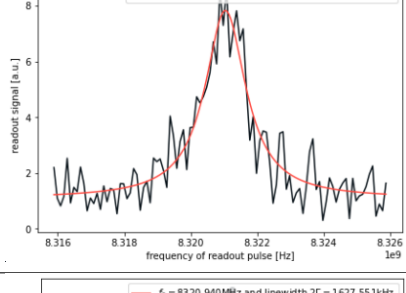
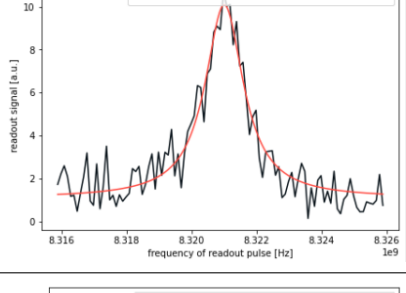
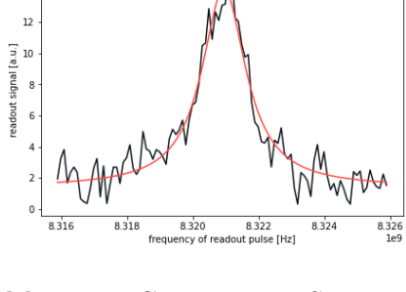
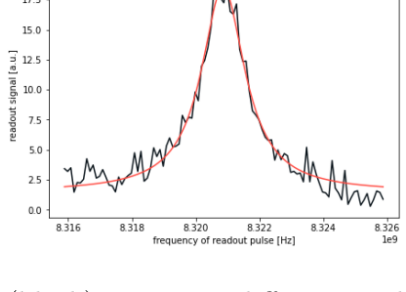
Data and fit	R^2	Data and fit	R^2
	0.251		0.383
	0.380		0.480
	0.654		0.759
	0.864		0.903
	0.924		0.961

Table B.2.: Cavity No.1: Spectroscopy data (black) at various different readout powers together with the coefficient of determination for the fitted Lorentzian (red).

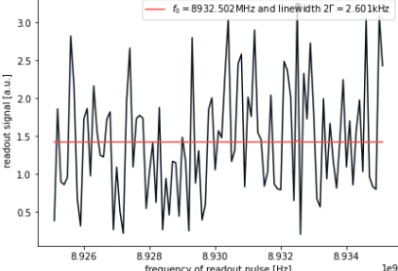
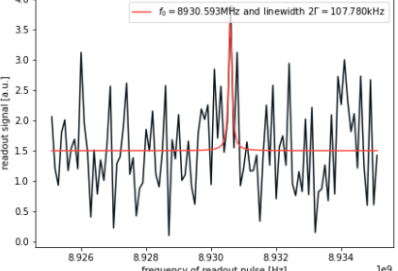
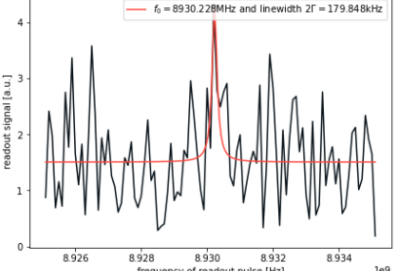
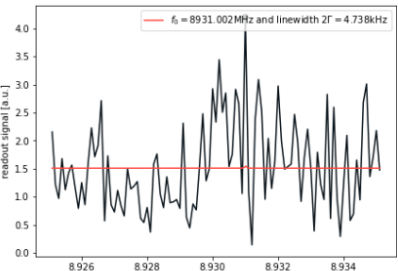
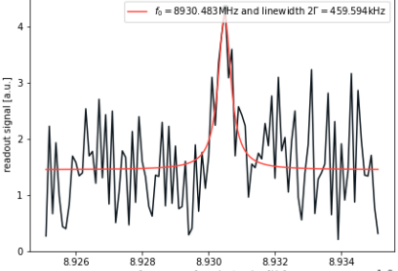
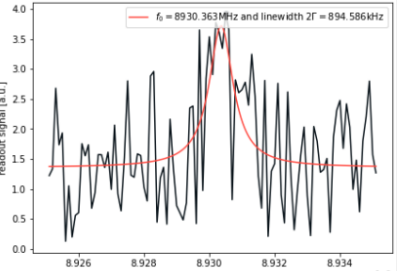
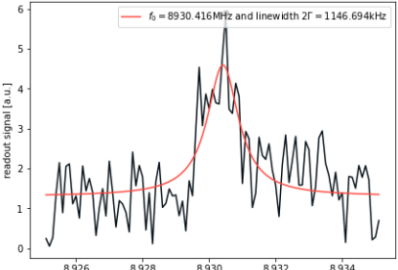
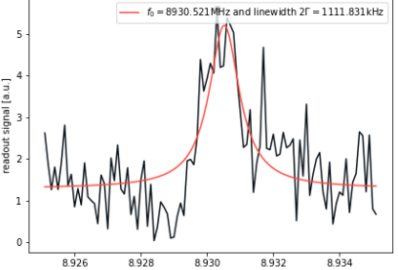
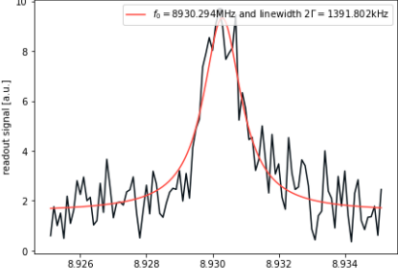
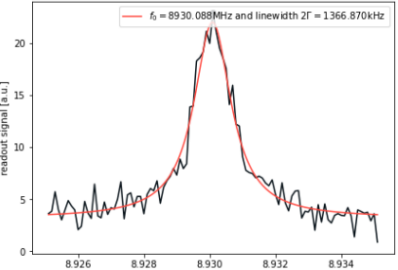
Data and fit	R^2	Data and fit	R^2
	0.055		0.107
	0.149		0.114
	0.313		0.325
	0.549		0.596
	0.838		0.956

Table B.3.: Cavity No.2: Spectroscopy data (black) at various different readout powers together with the coefficient of determination for the fitted Lorentzian (red).

B. Determining threshold values for the coefficient of determination

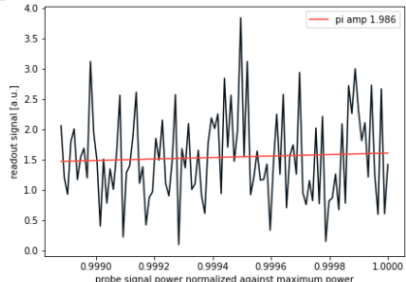
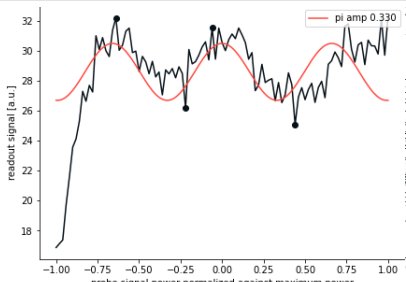
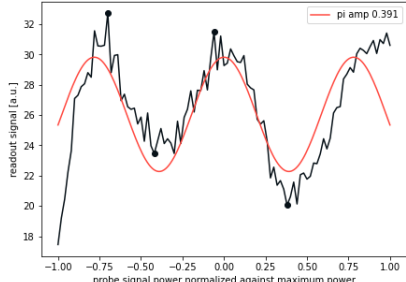
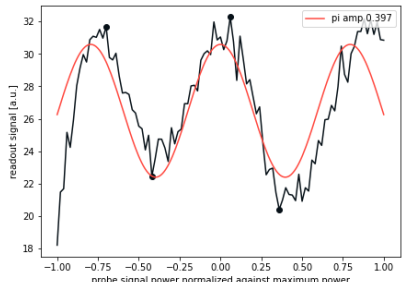
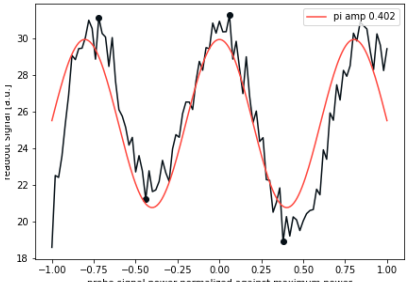
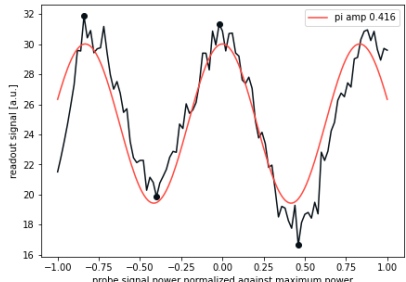
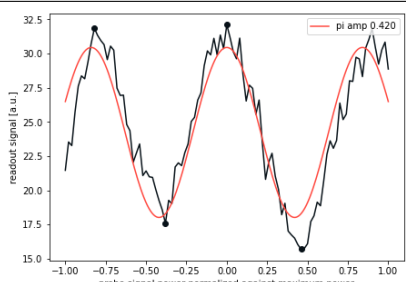
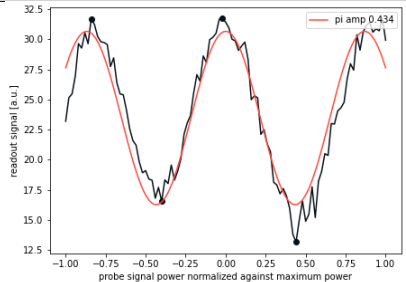
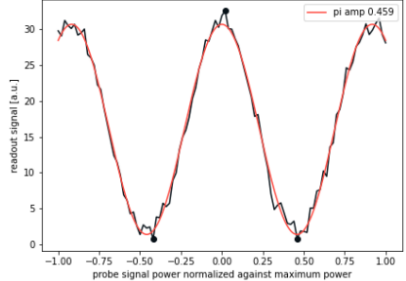
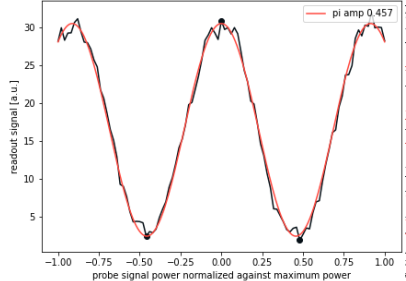
Data and fit	R^2	Data and fit	R^2
	0.003		0.219
	0.577		0.638
	0.756		0.818
	0.853		0.913
	0.993		0.994

Table B.4.: Rabi oscillation data (black) at different detunings between qubit and control pulse together with the coefficient of determination for the fit (red).

Standard parameter values

In this section we list all of the parameters that used at some point in the characterization procedure. All of these parameters can be changed via the GUI. They can be found in table C.1. A detailed description of the parameters function can be found in the GUI. The threshold values and parameters used in fine-tuning the peak detection algorithm that have already been by listed in appendix A and B are not listed here.

C. Standard parameter values

parameter name	value (unit)
ro_power	[-30, 13, 0.1] (dBm)
probe_power	[-30, 13, 0.1] (dBm)
ro_frequencies	[7.5e9, 9.5e9, 1e3] (Hz)
probe_frequencies'	[4e9, 8e9, 1e3] (Hz)
pulse_length	[1, 1e6, 1] (ns)
Rabi_power	[-1, 1, 0.001] (normalized power)
time	[1, 1e9, 1] (ns)
stepsize_cavity_spec_coarse	0.5e6 (Hz)
stepsize_cavity_spec_fine	0.1e6 (Hz)
stepsize_qubit_find	0.2e6 (Hz)
stepsize_qubit_spec	0.05e6 (Hz)
stepsize_ro_power	2 (dBm)
stepsize_probe_power	2 (dBm)
stepsize_Rabi	0.01 (normalized power)
ro_pulse_length	4000 (ns)
probe_pulse_length	10 (ns)
ef_probe_pulse_length	100 (ns)
T1_time	100e3 (ns)
stepsize_T1	0.2e3 (ns)
T2_time	100e3 (ns)
stepsize_T2	0.2e3 (ns)
T2_delta	500e3 (Hz)
T2_delta_difference	0.1 (percent of T2_delta)
T2_determine_delta_time	6e3 (ns)
stepsize_T2_determine_delta	0.05e3 (ns)
max_expected_chi	20e6 (Hz)
max_expected_alpha	150e6 (Hz)
naverage	3e3
intermediate_ro_power	$\frac{\text{ro_power}[0] + \text{ro_power}[1]}{2}$ (dBm)
intermediate_probe_power	$\frac{\text{probe_power}[0] + \text{probe_power}[1]}{2}$ (dBm)
probe_power_for_alpha	$\frac{\text{probe_power}[0] + 3 \times \text{probe_power}[1]}{4}$ (dBm)
alazar_acq_len	None
alazar_nsamples	None
pi_amp_treshold_lower	0.1 (normalized power)
pi_amp_treshold_upper	0.75 (normalized power)
max_repetitions_for_cav_spec	4
max_repetitions_for_decoherence_measurements	2
max_repetitions_for_Rabi_pulse_length_search	6
decay_time_multiple	6
savefig	True

Table C.1.: Values of the standard parameters that are used throughout the characterization procedure.

Example of log file

On the following pages an excerpt of an example log file can be found.

example_log_file.log

```

29.04 13:36:57 — Characterization procedure started with the following
    parameters:
  - cavity_resonance_warm: 8700000000.0
  - qubit_resonances_warm: [5200000000.0, 6400000000.0, 7000000000.0]
  - perform_alpha: True
  - perform_chi: True
  - perform_ef_Rabi: False
All Data is saved under: characterization_module\dummy_data.h5py
  - calibration_parameters: {'ro_power': [-30, 13, 0.1], 'probe_power':
    [-30, 13, 0.1], 'ro_frequencies': [7500000000.0, 9500000000.0,
    1000.0], 'probe_frequencies': [4000000000.0, 8000000000.0, 1000.0], '
    pulse_length': [1, 1000000.0, 1], 'Rabi_power': [-1, 1, 0.001], 'time
    ': [1, 1000000000.0, 1], 'stepsize_cavity_spec_coarse': 500000.0, '
    stepsize_cavity_spec_fine': 100000.0, 'stepsize_qubit_find': 200000.0,
    'stepsize_qubit_spec': 50000.0, 'stepsize_ro_power': 2, '
    stepsize_probe_power': 2, 'stepsize_Rabi': 0.01, 'ro_pulse_length':
    4000, 'probe_pulse_length': 10, 'ef_probe_pulse_length': 100, 'T1_time
    ': 100000.0, 'stepsize_T1': 200.0, 'T2_time': 100000.0, 'stepsize_T2':
    200.0, 'T2_delta': 500000.0, 'T2_delta_difference': 0.1, '
    T2_determine_delta_time': 6000.0, 'stepsize_T2_determine_delta': 50.0,
    'max_expected_chi': 20000000.0, 'max_expected_alpha': 150000000.0, '
    naverage': 3000.0, 'intermediate_ro_power': -8.5, '
    intermediate_probe_power': -8.5, 'probe_power_for_alpha': 2.25, '
    alazar_acq_len': None, 'alazar_nsamples': None, 'R^2_threshold_cavity
    ': 0.4, 'R^2_optimized_cavity_threshold': 0.9, 'R^2_Rabi_threshold':
    0.15, 'R^2_optimized_Rabi_threshold': 0.9, 'R^2_threshold_qubit':
    0.5, 'R^2_T1_threshold': 0.45, 'R^2_T2_threshold': 0.45, 'R^2
    _T2_echo_threshold': 0.45, 'pi_amp_treshold_lower': 0.1, '
    pi_amp_treshold_upper': 0.75, 'max_repetitions_for_cav_spec': 4, '
    max_repetitions_for_decoherence_measurements': 2, '
    max_repetitions_for_Rabi_pulse_length_search': 6, 'decay_time_multiple
    ': 6, 'savefig': True}
29.04 13:36:57 — STEP 0: FINDING THE CAVITY RESONANCE —
29.04 13:36:57 — Cavity Spectroscopy (cavity_find) with the following
    parameters:
  - ro_freqs: [8650000000Hz, 8900000000Hz] in steps of 500000Hz
  - ro_power: -8.50dBm

```

D. Example of log file

```
– ro_pulse_length: 4000ns
29.04 13:36:57 — Measurement performed succesfully. Data under:
    dummy_cavity_find
29.04 13:36:58 — The analysis yields:
– center frequency: 8930102268Hz
– linewidth: 1674166Hz
29.04 13:36:58 — Cavity Spectroscopy (cavity_spec) with the following
    parameters:
– ro_freqs: [8925080000Hz, 8935124000Hz] in steps of 100440Hz
– ro_power: -8.50dBm
– ro_pulse_length: 4000ns
29.04 13:36:58 — Measurement performed succesfully. Data under:
    dummy_cavity_spec
29.04 13:36:59 — The analysis yields:
– center frequency: 8929951000Hz
– linewidth: 1568000Hz
– R^2: 0.98120685914961
29.04 13:36:59 — DECISION: This R^2 lies above the threshold to count as
    a Lorentzian lineshape. Therefore the cavity was found.
29.04 13:36:59 — Cavity Spectroscopy (optimize_ro_power) with the
    following parameters:
– ro_freqs: [8925247000Hz, 8934655000Hz] in steps of 94080Hz
– ro_power: range(-30, 13, 2)dBm
– ro_pulse_length: 4000Hz
29.04 13:36:59 — Measurement performed succesfully. Data under:
    dummy_ro_opt
29.04 13:37:02 — From the analysis we determined the optimal ro_power:
    -2.7dBm
Perform cavity spectroscopy at the optimal readout power to determine the
    cavity's linewidth and resonance frequency.
29.04 13:37:02 — Cavity Spectroscopy (cavity_spec) with the following
    parameters:
– ro_freqs: [8925247000Hz, 8934655000Hz] in steps of 94080Hz
– ro_power: -2.70dBm
– ro_pulse_length: 4000ns
29.04 13:37:02 — Measurement performed succesfully. Data under:
    dummy_cavity_spec
29.04 13:37:02 — The analysis yields:
– cavity resonance frequency: 8929951000Hz
– cavity linewidth: 1568000Hz
– readout level: 31.712316407458097
– R^2: 0.98120685914961
29.04 13:37:02 — ————— FINDING QUBIT CANDIDATES —————
29.04 13:37:02 — Do broad qubit spectroscopy to find potential qubit dips
.
29.04 13:37:02 — Qubit Spectroscopy (qubit_find) with the following
    parameters:
– probe_freqs: [4200000000Hz, 8000000000Hz] in steps of 200000Hz
– probe_pulse: 10ns length at -8.50dBm
– readout_pulse: 4000ns at 8929951000Hz and -2.700000dBm
29.04 13:37:02 — Measurement performed succesfully. Data under:
    dummy_qubit_find
```

Example of PDF report file

On the following pages an excerpt of an example PDF report can be found.

Dummy_characterization

Data under:

characterization_module\dummy_data.h5py

Logfiles under:

characterization_module\Logfiles\Dummy_characterization.log

Cavity spectroscopy and Readout optimization

Instance of cavity class:

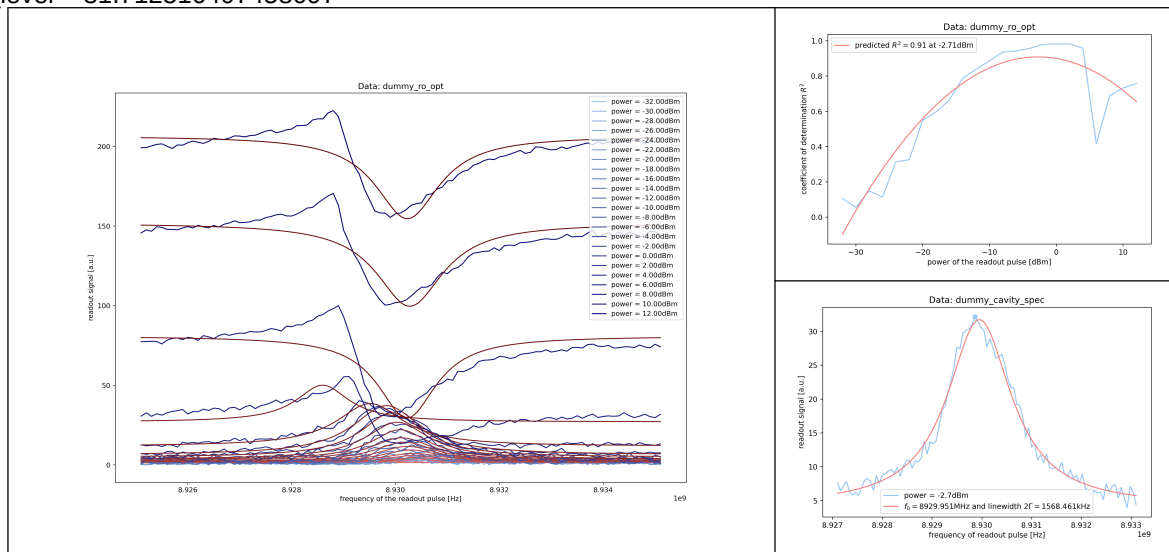
- resonance_frequency = 8929951000.0Hz

- linewidth = 1568000.0Hz

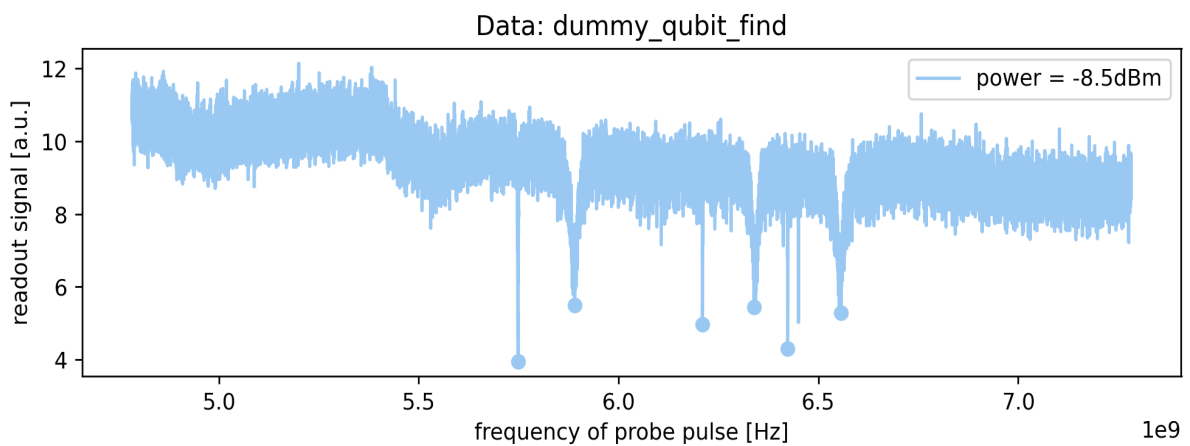
- ro_pulse_length = 4000ns

- ro_power = -2.7dBm

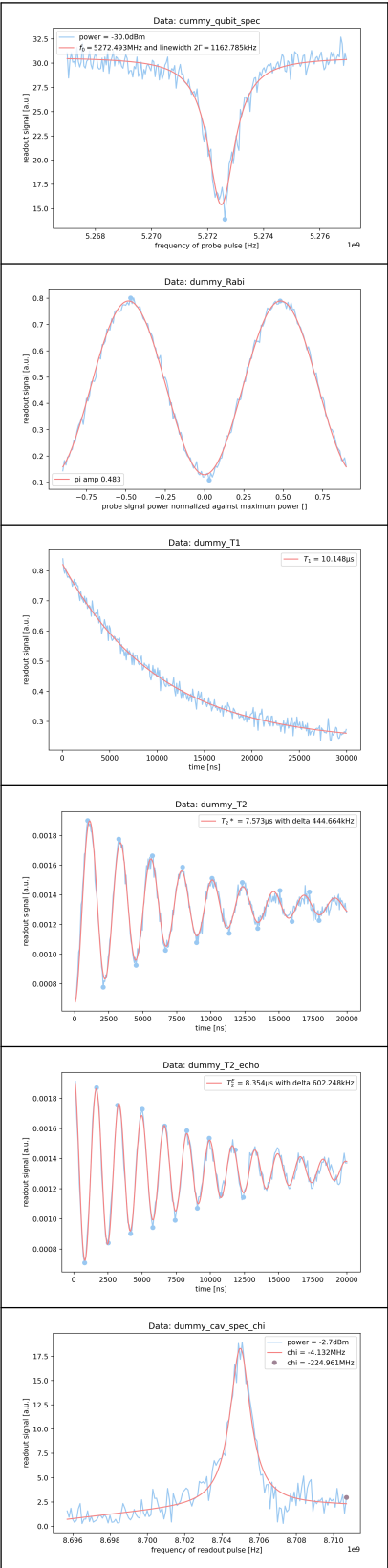
- ro_level = 31.712316407458097



Finding qubits



Characterizing Qubit #1



Instance of qubit class:

ge Transition:

- resonance_frequency = 5272548000.0Hz
- linewidth = 1163000.0Hz
- probe_pulse_length = 10ns
- probe_power = None
- pi_amp = 0.483
- T1 = 10148.0ns
- T2 = 7573.0ns
- T2_echo = 8354.0ns
- chi = -224961000.0Hz

ef Transition:

- alpha = 92000.0Hz
- ef_resonance_frequency = None
- ef_probe_pulse_length = None
- ef_rabi_amp = None

Characterization Info:

- status = fully_characterized
- ID = 1
- figures = ['qubit_spec', 'Rabi', 'T2', 'T2_echo', 'T1', 'chi', 'alpha']

